



Boeing Contributions to the Scatter Reduction Working Group of the 8th Drag Prediction Workshop

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Boeing Commercial Airplanes
Flight Sciences - Aerodynamics





- **Flow Solvers**

- OVERFLOW
- GGNS-T1
- FELight

- **Test Cases**

- Test Case 1a: ONERA OAT15A Airfoil
- Test Case 1b: Joukowski Airfoil
- Test Case 1c: ONERA OAT15A Airfoil – Part Duex
- Test Case 1d/1e: DPW-III Case 2 Wing Only (W1/W2)
- Test Case 2a/2b: NASA CRM Wing/Body

- **Results**

- Force/Moments

- **Conclusions**



Flow Solvers Utilized



- **OVERFLOW (v2.4d)**

- NASA developed overset grid solver
- SA turbulence model (SA-neg) with many corrections (RC, CC, QCR2000, etc.)
- HLLC++ upwind scheme (IRHS=6) with Improved SSOR algorithm (Implicit BCs + Global Linear Solve) (ILHS=26)

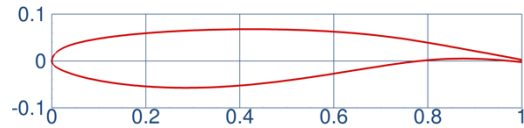
- **GGNS-T1**

- Boeing developed finite element solver for fixed grid or
- SUPG, piecewise-linear, globally continuous finite elements with residual-based stabilization
- Shock capturing/artificial dissipation, Newton's method, fully coupled turbulence equations
- SA, SA-QCR2000, SARC-QCR2000
- Grid adaptation (EPIC) (tetrahedral cells)

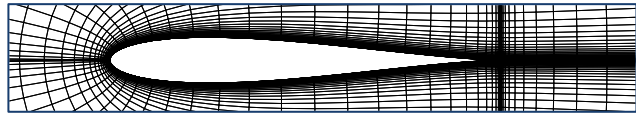
- **FELight**

- Boeing developed finite element solver
- SUPG, 2nd-order accurate in space for all equations (fully coupled turbulence equation), BFD2 and SDIRK for temporal disc.
- Shock capturing, Newton's method with preconditioned GMRes
- SA, SA-QCR2000, SARC-QCR2000

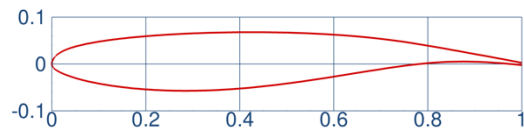
- 1a: ONERA OAT15A Transonic Airfoil



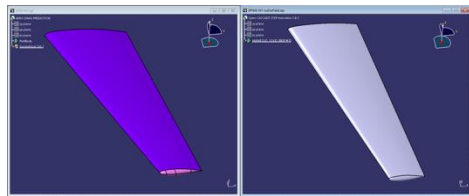
- 1b: Joukowski Airfoil (Subsonic)



- 1c: ONERA OAT15A (Transonic)



- 1d/1e: DPW-III W1/W2 Wings



- 2a/2b: NASA CRM Wing/Body



Grids:	OVERFLOW	FELight	GGNS
Cadence_Grids.REV01/Structured	X		
Custom* User Structured Grid	X		
Custom Grid Adaptation			X

Grids:	OVERFLOW	FELight	GGNS
Python CLASSIC Grid	X	X	

Grids:	OVERFLOW	FELight	GGNS
Helden_Grids.REV02		X	
Cadence_Grids.REV02/Unstructured		X	
Cadence_Grids.REV02/Structured	X	X	

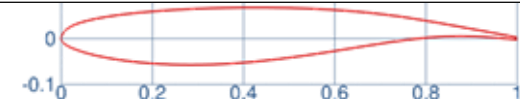
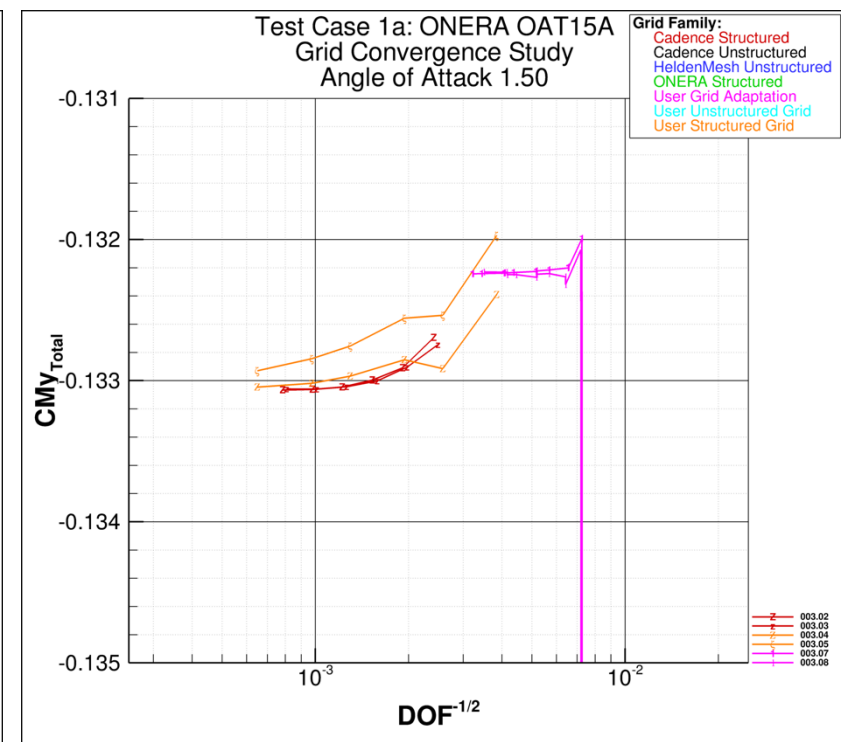
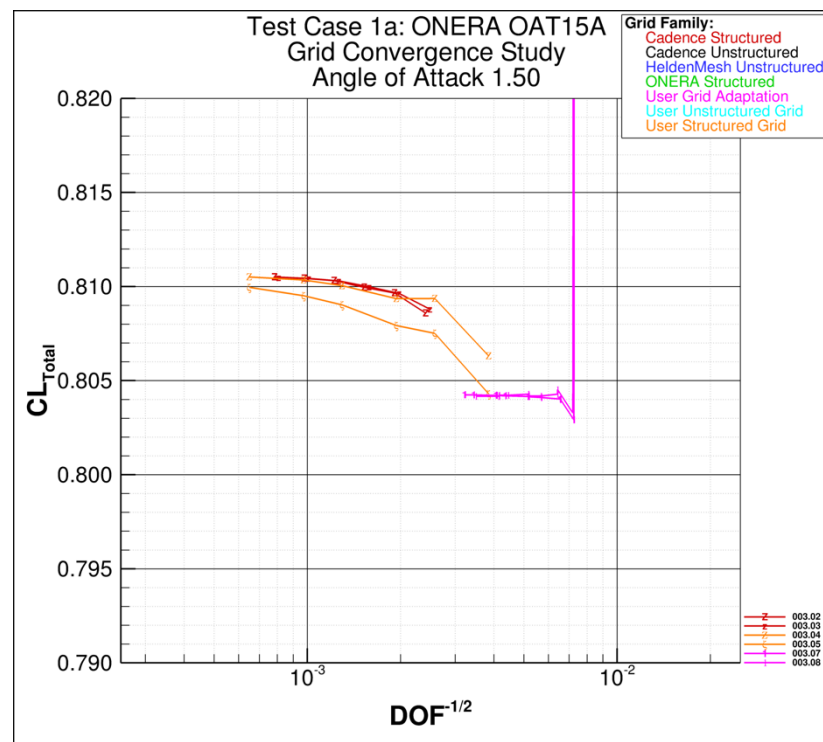
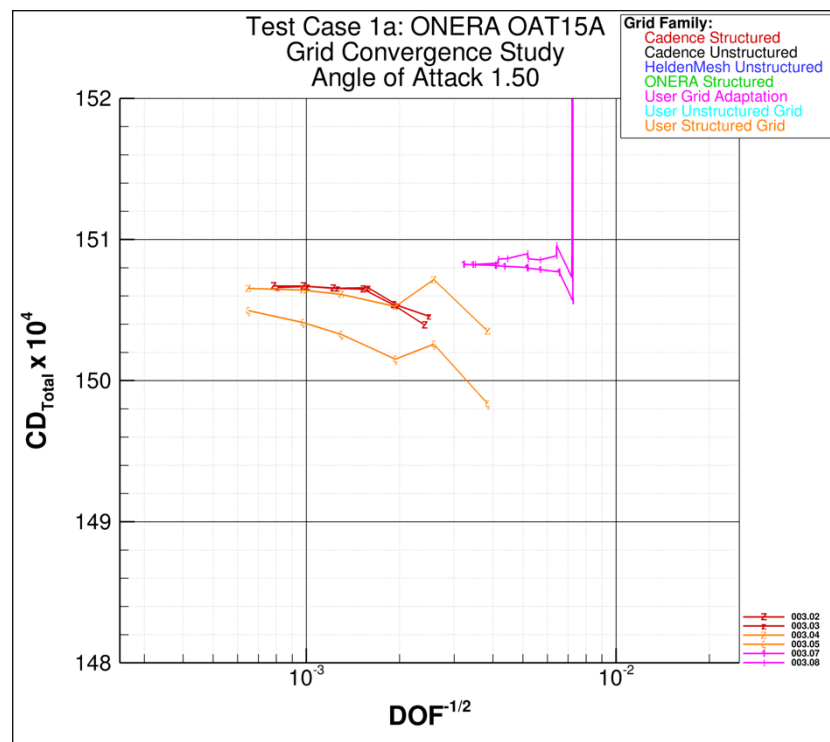
Grids	OVERFLOW	FELight	GGNS
Cadence_HexTrex_TetFF.REV00		X	X
Cadence_HexTrex_VoxelFF.REV00		X	X
HeldenMesh_Grids.REV00		X	X
Custom Grid Adaptation			X
Cadence_HexTrex_TetFF.REV01		X	
Cadence_HexTrex_VoxelFF.REV01		X	

Grids	OVERFLOW	FELight	GGNS
Cadence_Grids.REV00		X	
HeldenMesh_Grids.REV00		X	
Ames_Grids.REV00	X		

• ONERA OAT15A Airfoil

- Good force/moment grid convergence
- Some small variation between solvers

Grids:	OVERFLOW	FELight	GGNS
Cadence_Grids.REV01/Structured	X		
Custom* User Structured Grid	X		
Custom Grid Adaptation			X



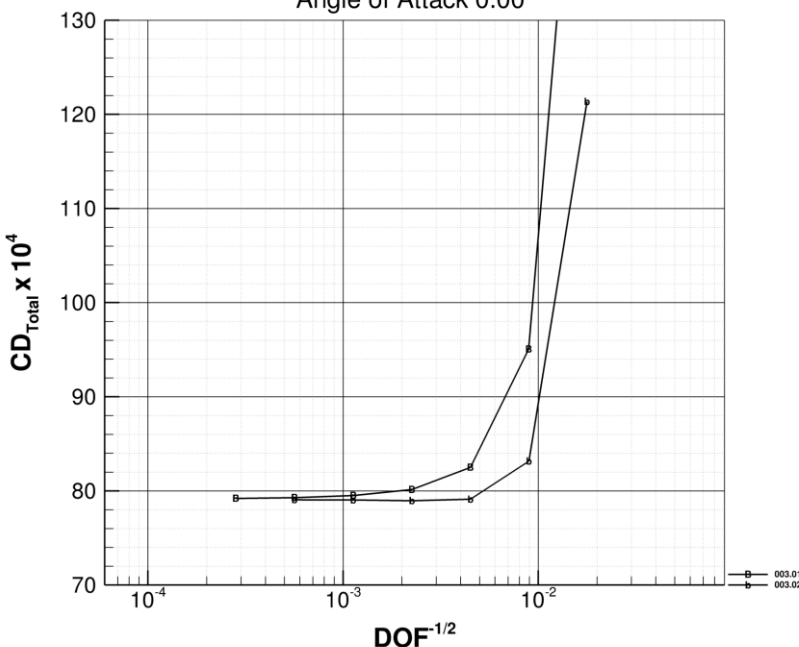
• Joukowski Airfoil

- Good force/moment grid convergence
- Very small variation between solvers
- Common grid, farfield, and problem description helped!

Grids:	OVERFLOW	FELight	GGNS
Python CLASSIC Grid	X	X	

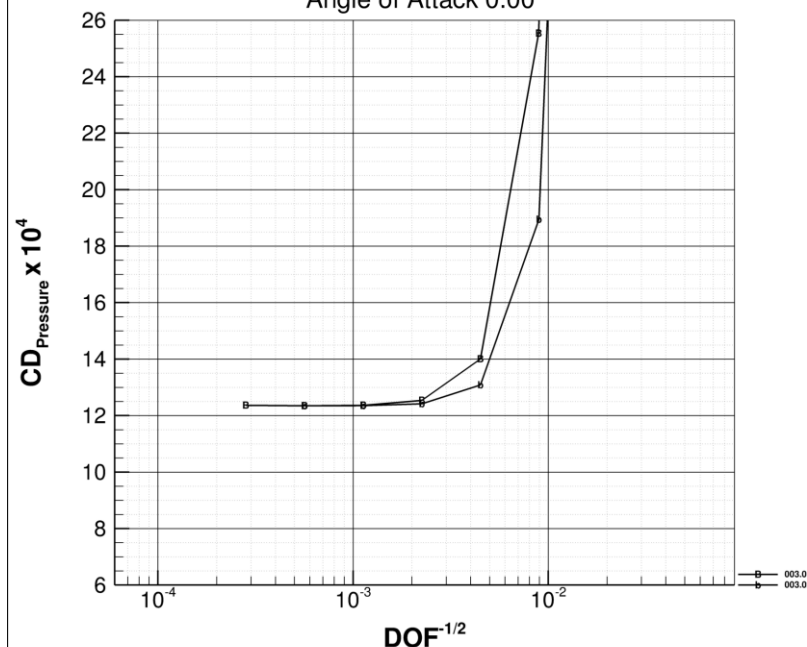
Test Case 1b: Joukowski Airfoil
Grid Convergence Study
Angle of Attack 0.00

Grid Family:
Python Mesh Family



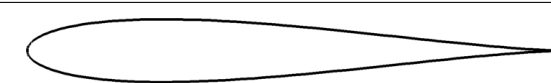
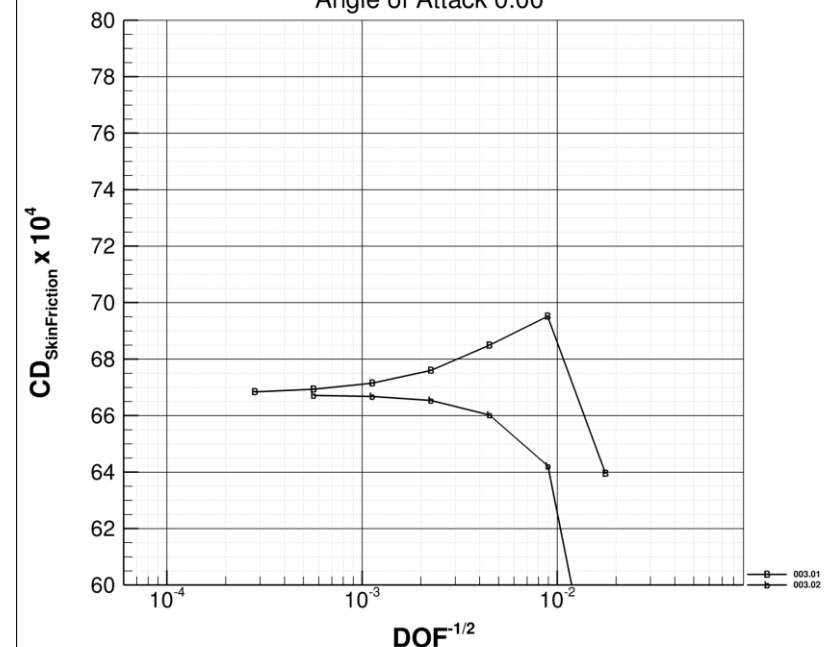
Test Case 1b: Joukowski Airfoil
Grid Convergence Study
Angle of Attack 0.00

Grid Family:
Python Mesh Family



Test Case 1b: Joukowski Airfoil
Grid Convergence Study
Angle of Attack 0.00

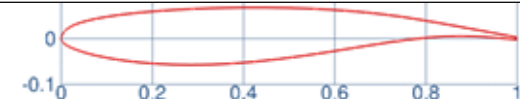
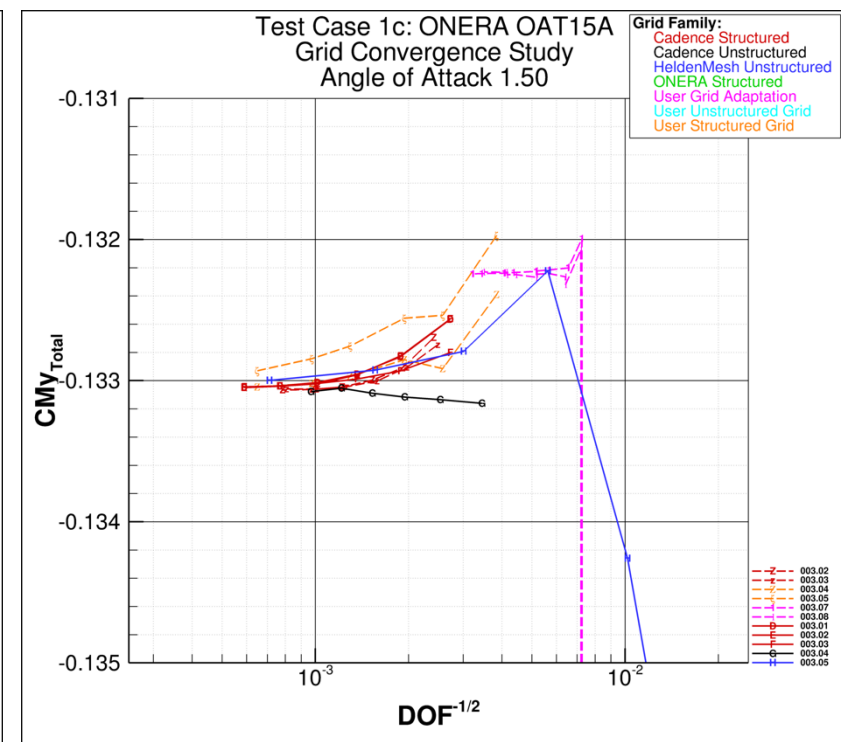
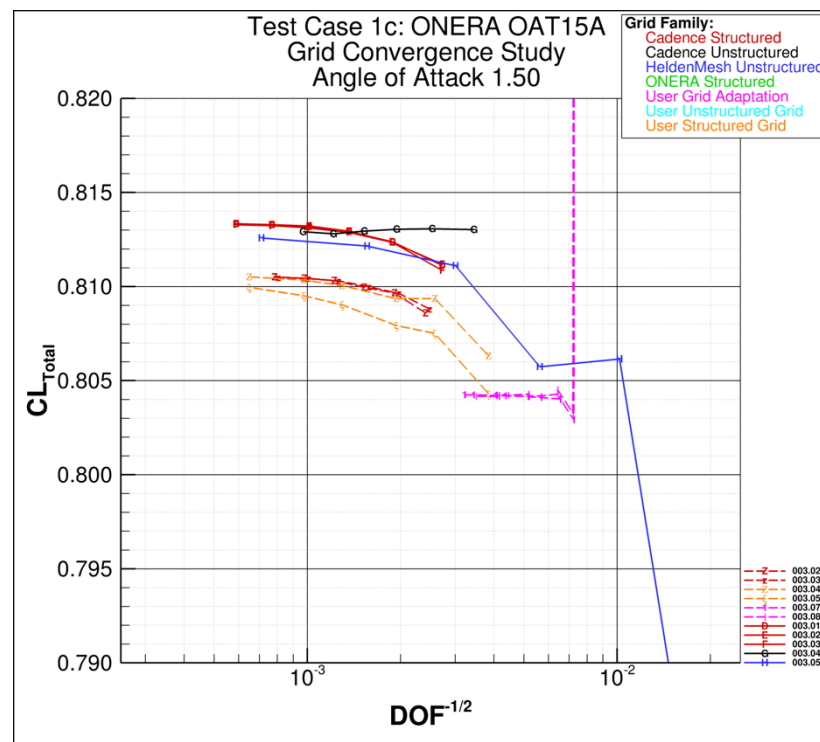
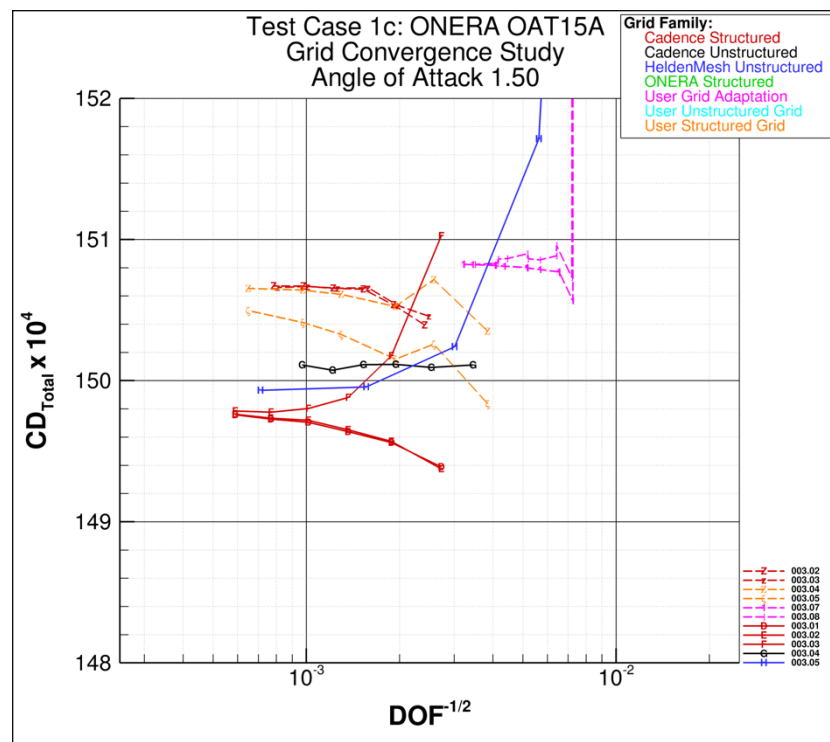
Grid Family:
Python Mesh Family



• ONERA OAT15A Airfoil

- Improved agreement – Reduced scatter!
 - Common inputs to the problem between solvers
 - Common farfield distance between grids

Grids:	OVERFLOW	FELight	GGNS
Holden_Grids.REV02		X	
Cadence_Grids.REV02/Unstructured		X	
Cadence_Grids.REV02/Structured	X	X	



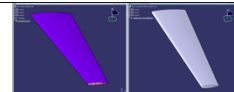
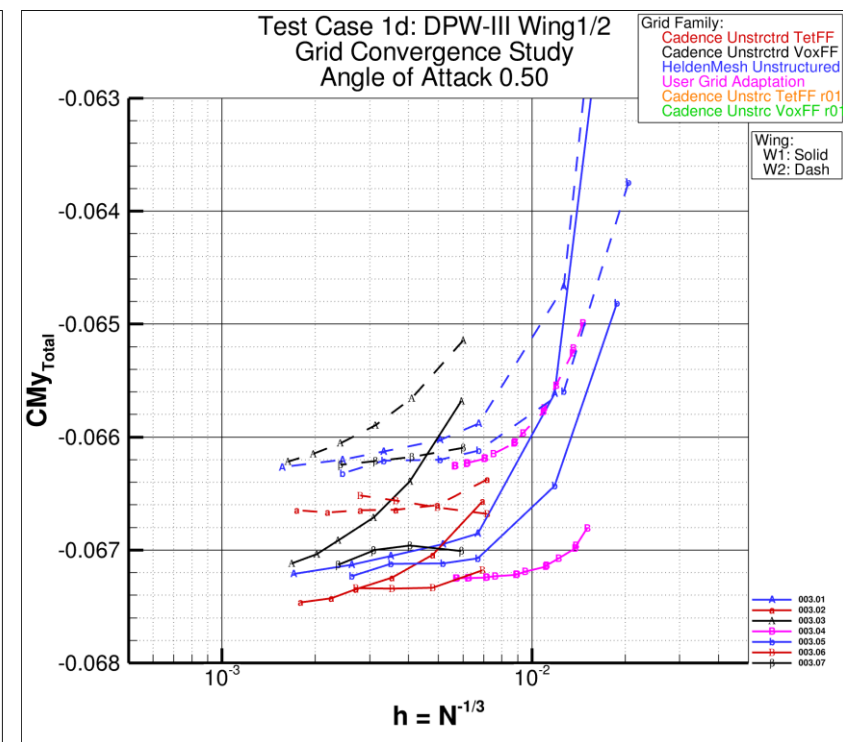
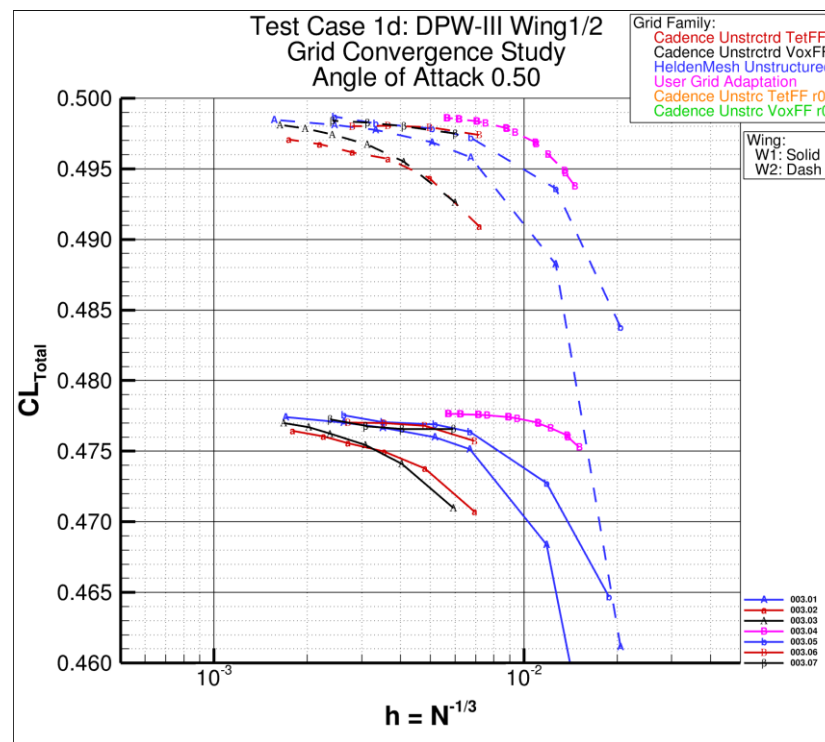
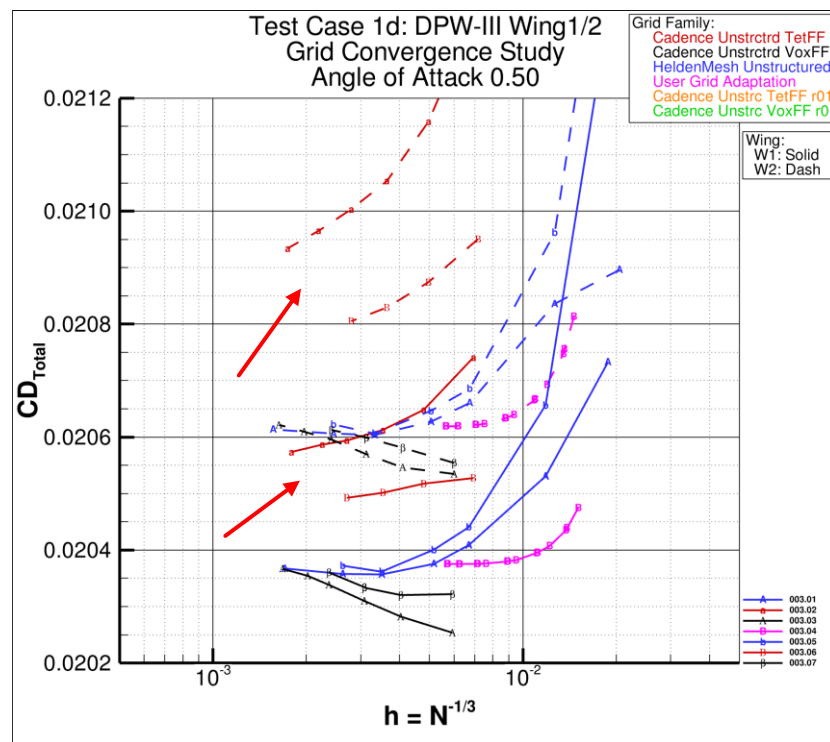


Results: Test Case 1d

• DPW-III: Wing-Only Increment Test Case

- Grid convergence not yet demonstrated
- Voxel FF(black), Helden(blue), and Adapted(pink) match
- Tet Farfield (red) deviates from other results for multiple solvers

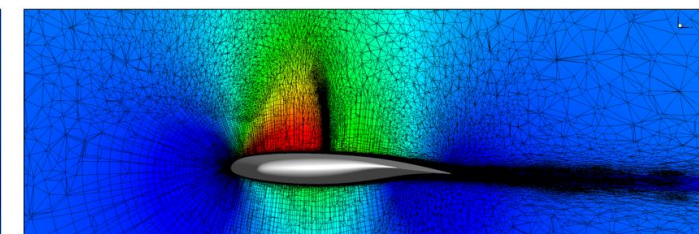
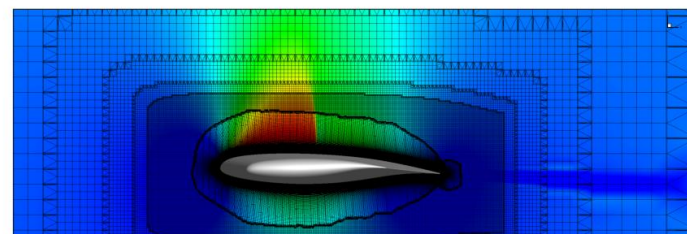
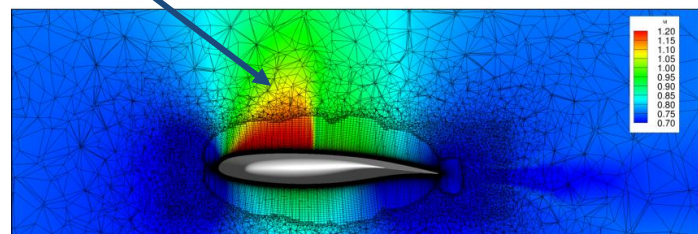
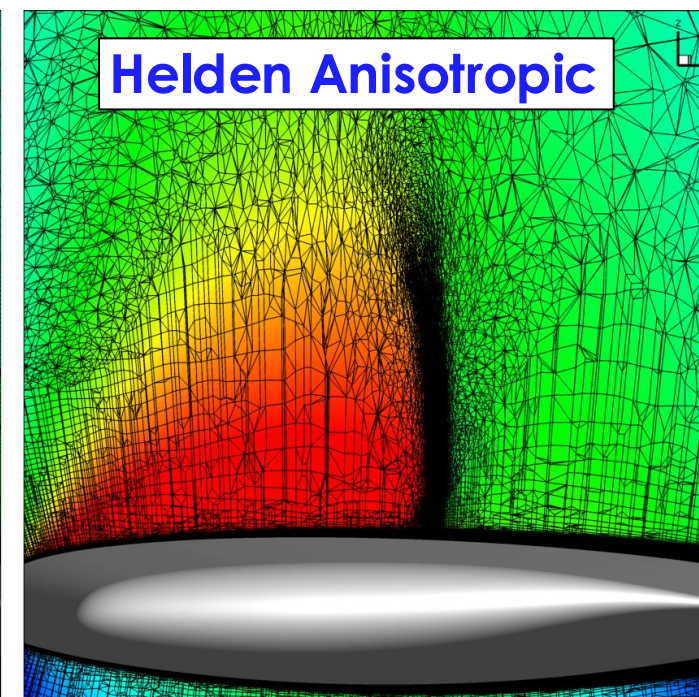
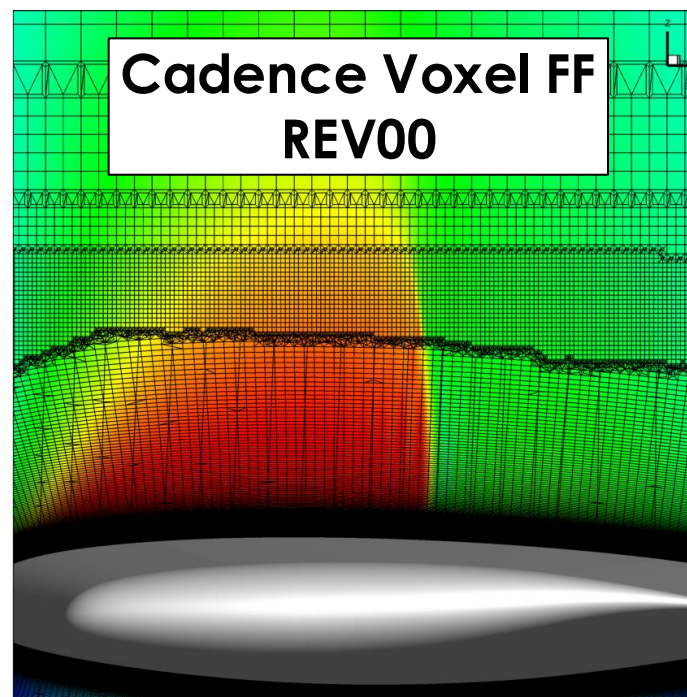
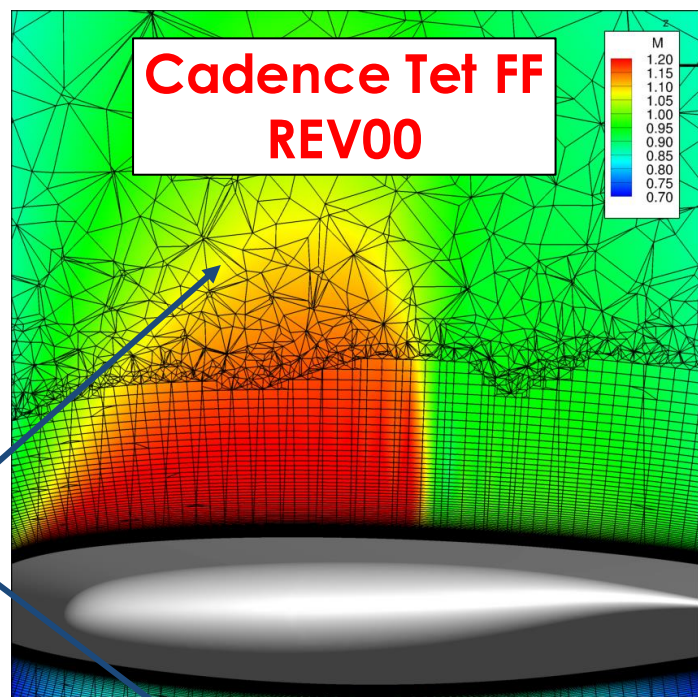
Grids	OVERFLOW	FELight	GGNS
Cadence_HexTrex_TetFF.REV00		X	X
Cadence_HexTrex_VoxelFF.REV00		X	X
HeldenMesh_Grids.REV00		X	X
Custom Grid Adaptation			X



- **Supersonic region and shock are not fully within near body mesh refinement**
 - As mesh is refined, the refinement only improves this area very slowly

Mach Contours
W1 | $\alpha = 0.5^\circ$

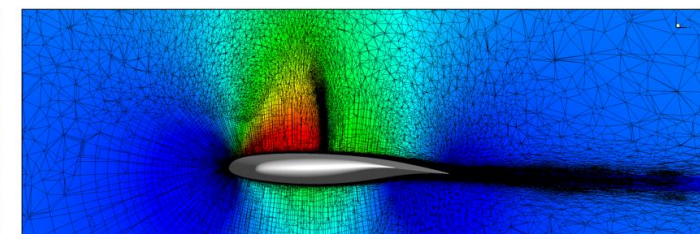
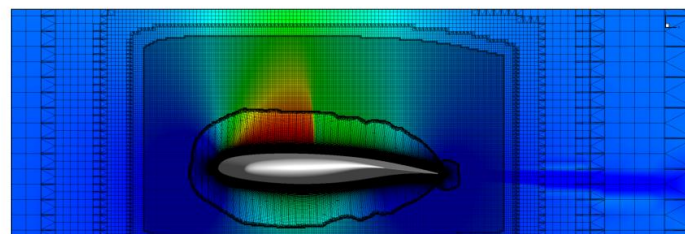
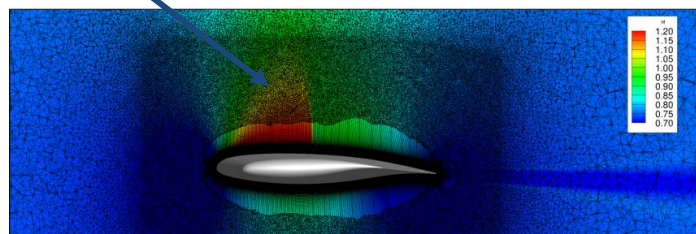
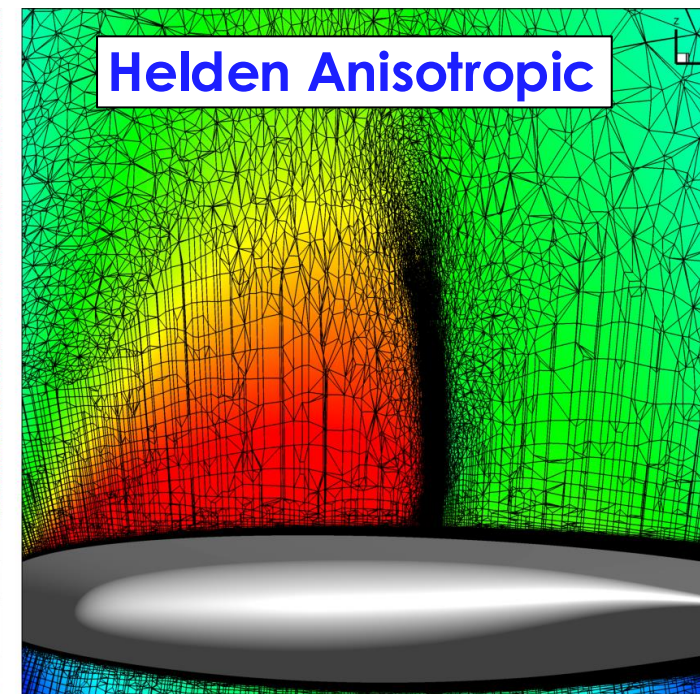
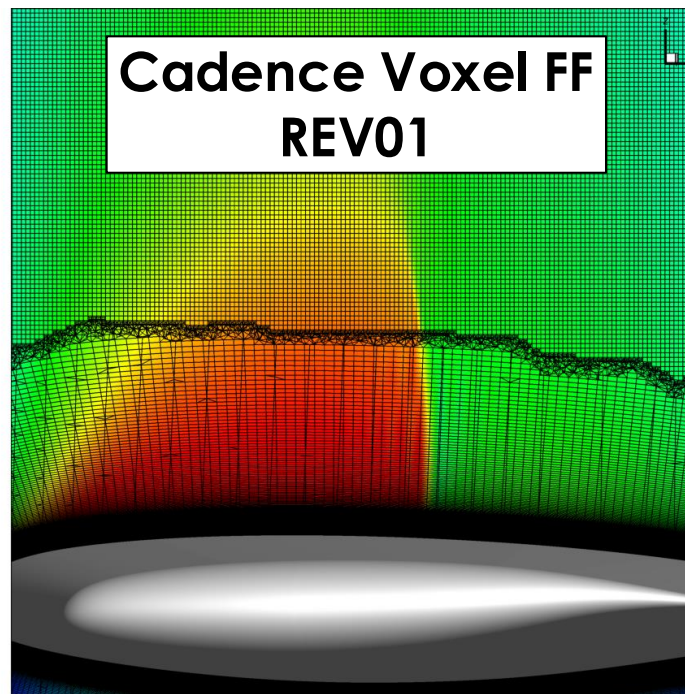
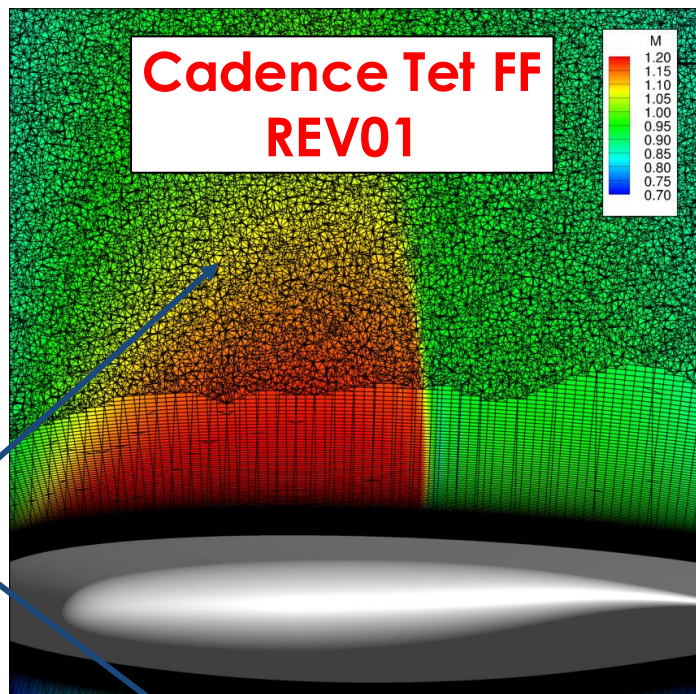
Refinement
limited



- Revised Tet and Voxel farfield meshes have larger near-body refinement region
 - Captures the upper surface velocity gradients much more accurately

Mach Contours
W1 | $\alpha = 0.5^\circ$

Refinement
expanded



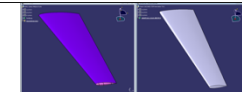
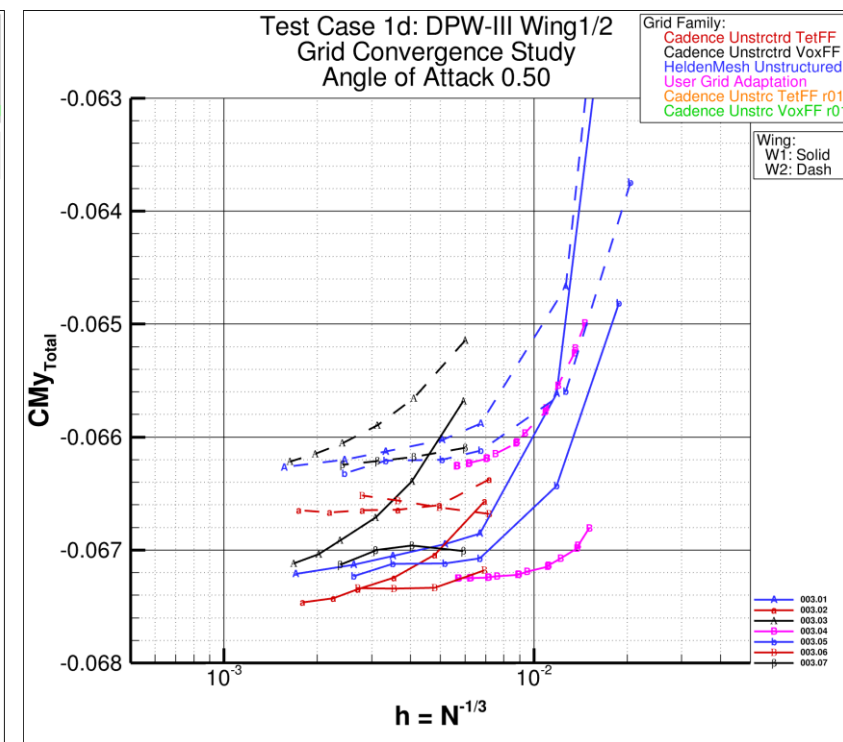
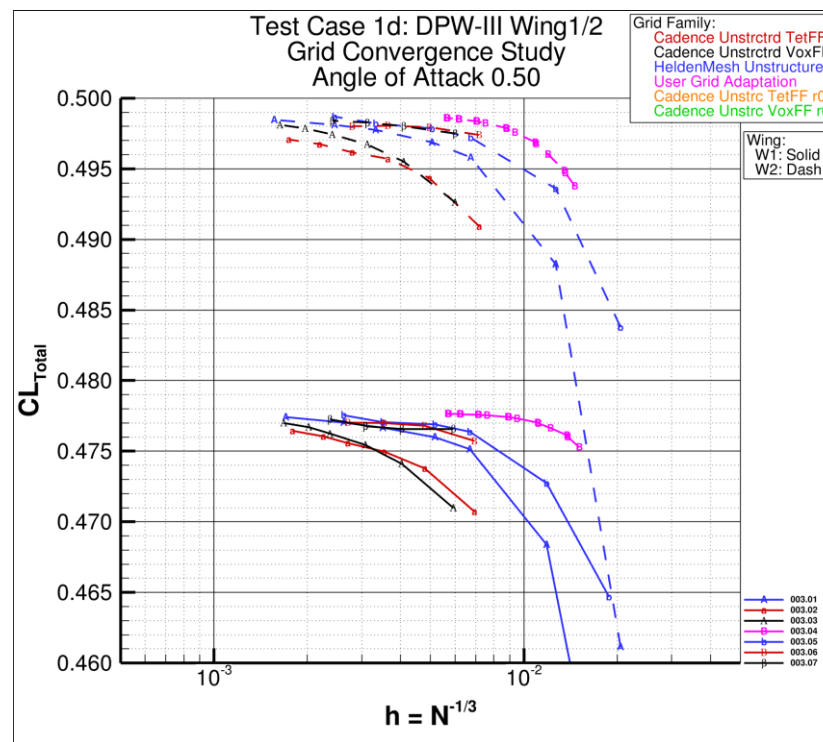
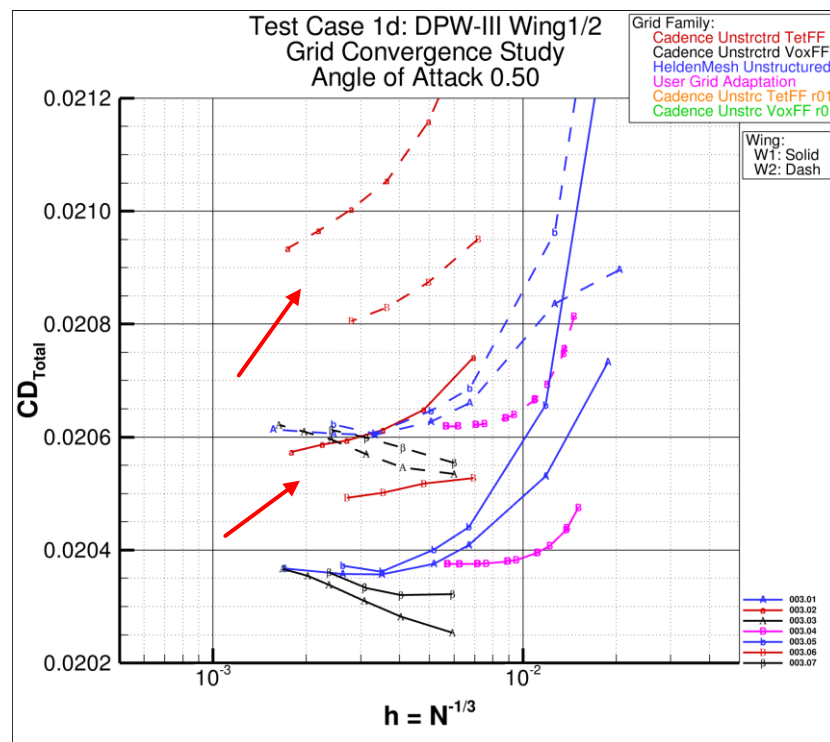


Results: Test Case 1d

• DPW-III: Wing-Only Increment Test Case

- Grid convergence not yet demonstrated
- Voxel FF(black), Helden(blue), and Adapted(pink) match
- Tet Farfield (red) deviates from other results

Grids	OVERFLOW	FElight	GGNS
Cadence_HexTrex_TetFF.REV00		X	X
Cadence_HexTrex_VoxelFF.REV00		X	X
HeldenMesh_Grids.REV00		X	X
<i>Custom Grid Adaptation</i>			X
Cadence_HexTrex_TetFF.REV01		X	
Cadence_HexTrex_VoxelFF.REV01		X	



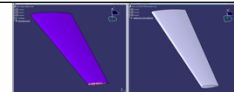
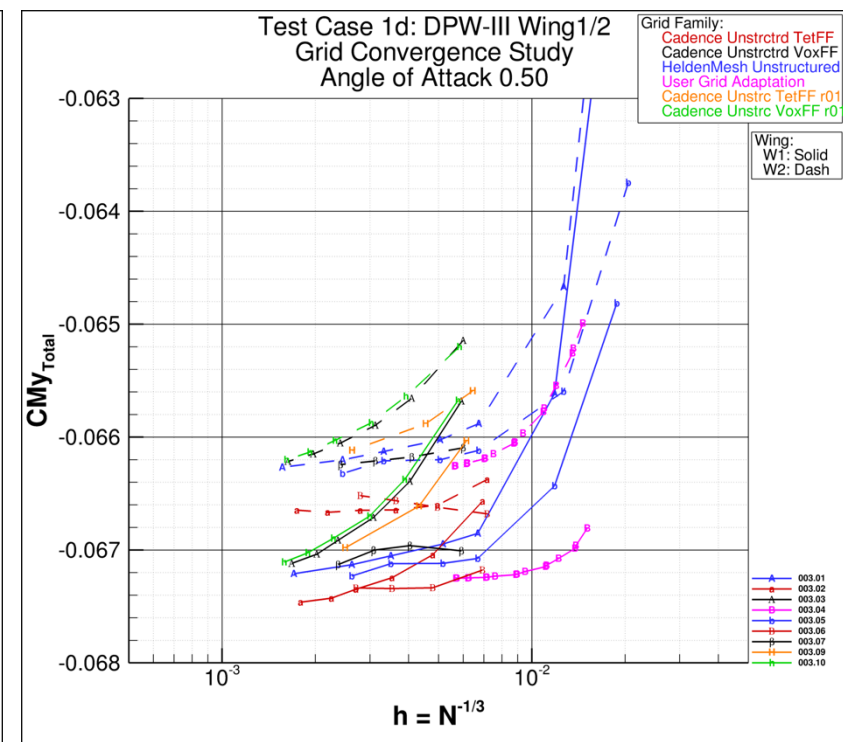
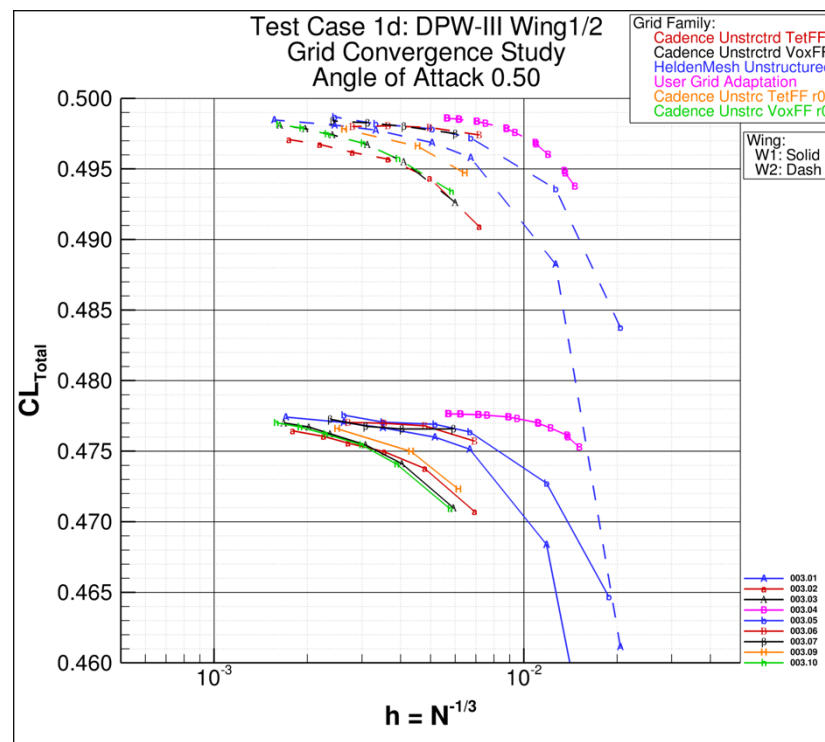
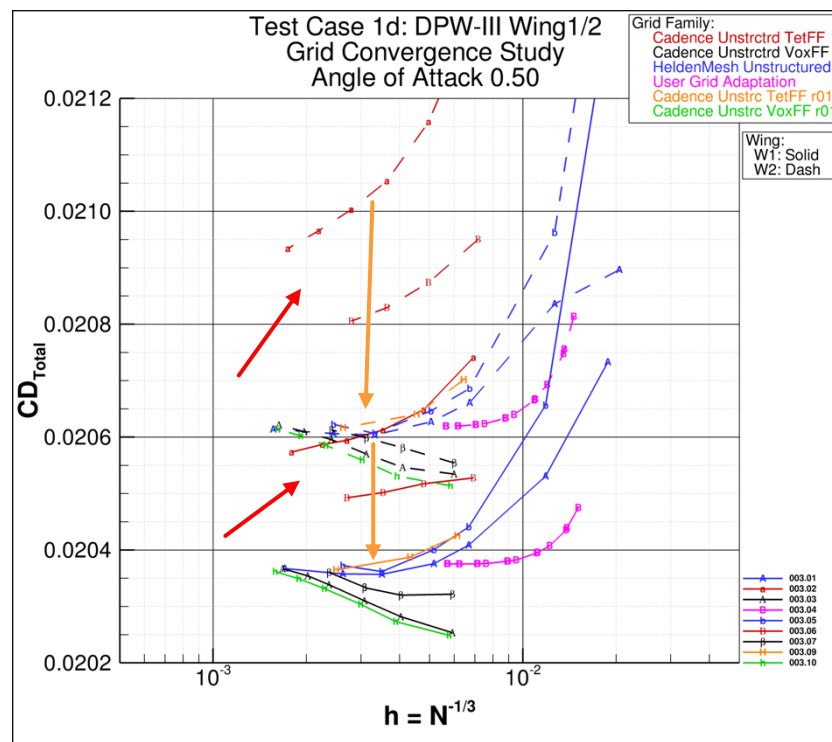


Results: Test Case 1d

• DPW-III: Wing-Only Increment Test Case

- Grid convergence not yet demonstrated
- Voxel FF(black), Helden(blue), and Adapted(pink) match
- REV01 Tet Farfield (orange) shows good agreement

Grids	OVERFLOW	FElight	GGNS
Cadence_HexTrex_TetFF.REV00		X	X
Cadence_HexTrex_VoxelFF.REV00		X	X
HeldenMesh_Grids.REV00		X	X
<i>Custom Grid Adaptation</i>			X
Cadence_HexTrex_TetFF.REV01		X	
Cadence_HexTrex_VoxelFF.REV01		X	



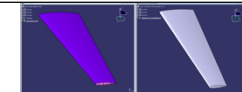
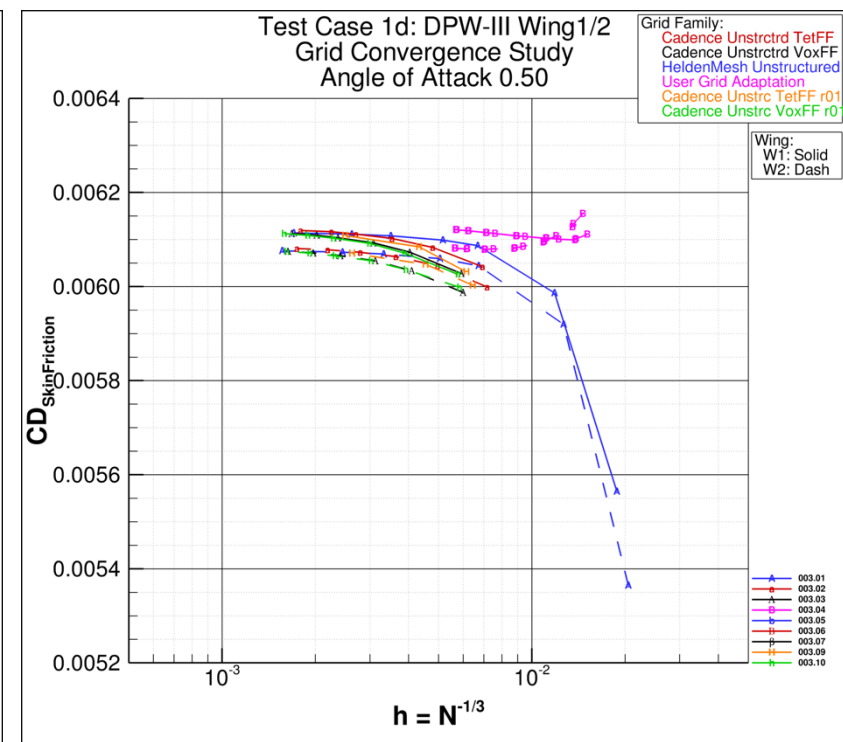
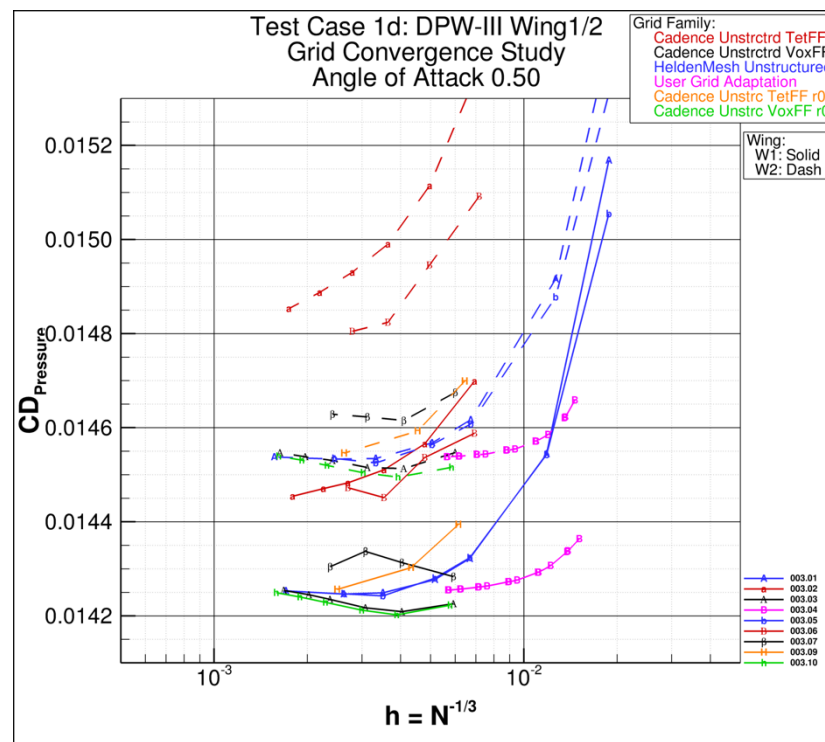
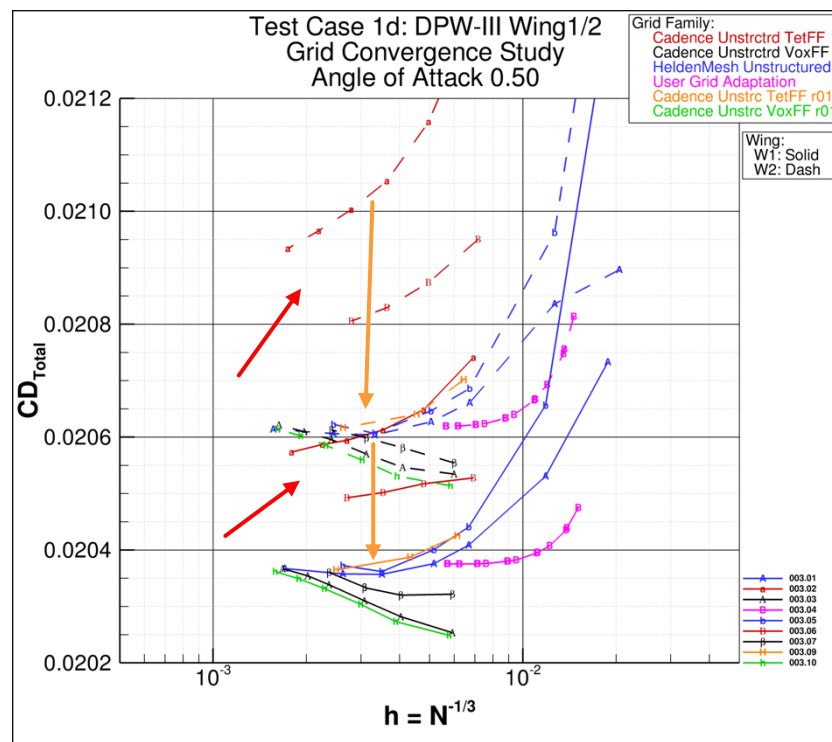


Results: Test Case 1d

• DPW-III: Wing-Only Increment Test Case

- Grid convergence not yet demonstrated
- Voxel FF(black), Helden(blue), and Adapted(pink) match
- REV01 Tet Farfield (orange) shows good agreement

Grids	OVERFLOW	FELight	GGNS
Cadence_HexTrex_TetFF.REV00		X	X
Cadence_HexTrex_VoxelFF.REV00		X	X
HeldenMesh_Grids.REV00		X	X
<i>Custom Grid Adaptation</i>			X
Cadence_HexTrex_TetFF.REV01		X	
Cadence_HexTrex_VoxelFF.REV01		X	



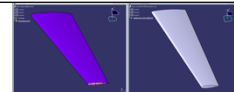
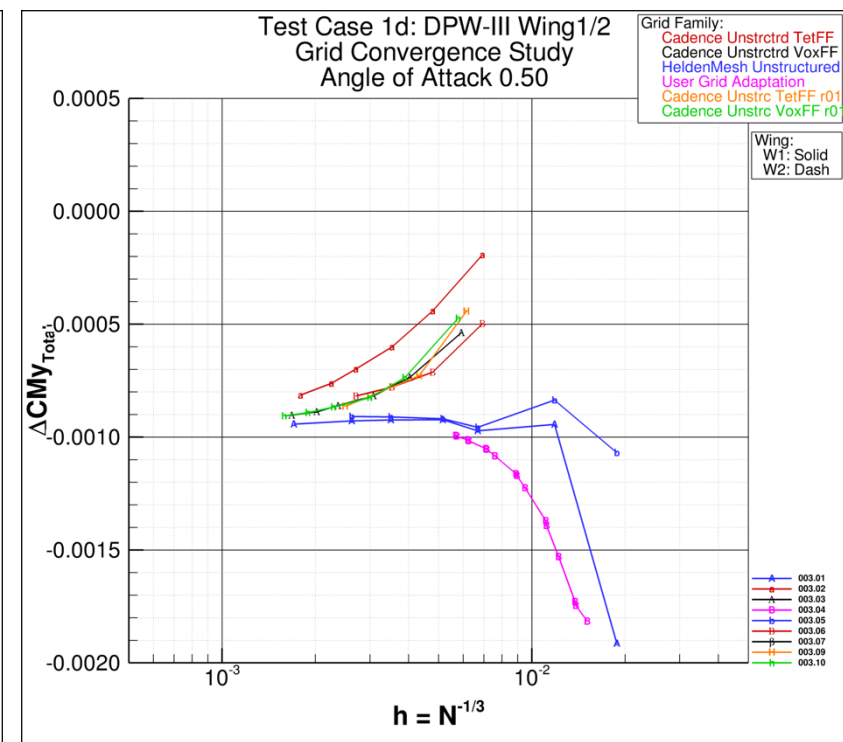
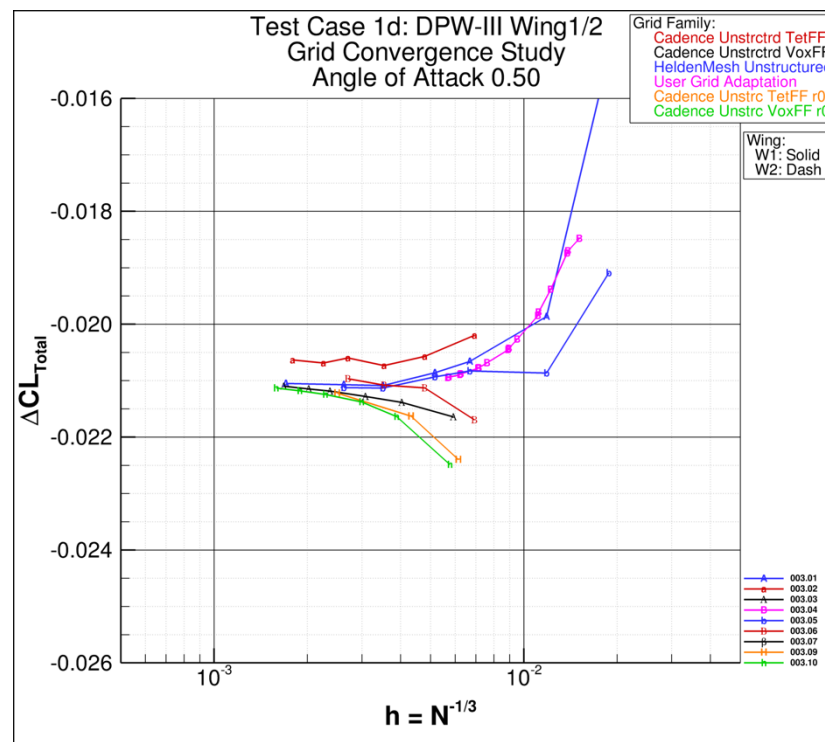
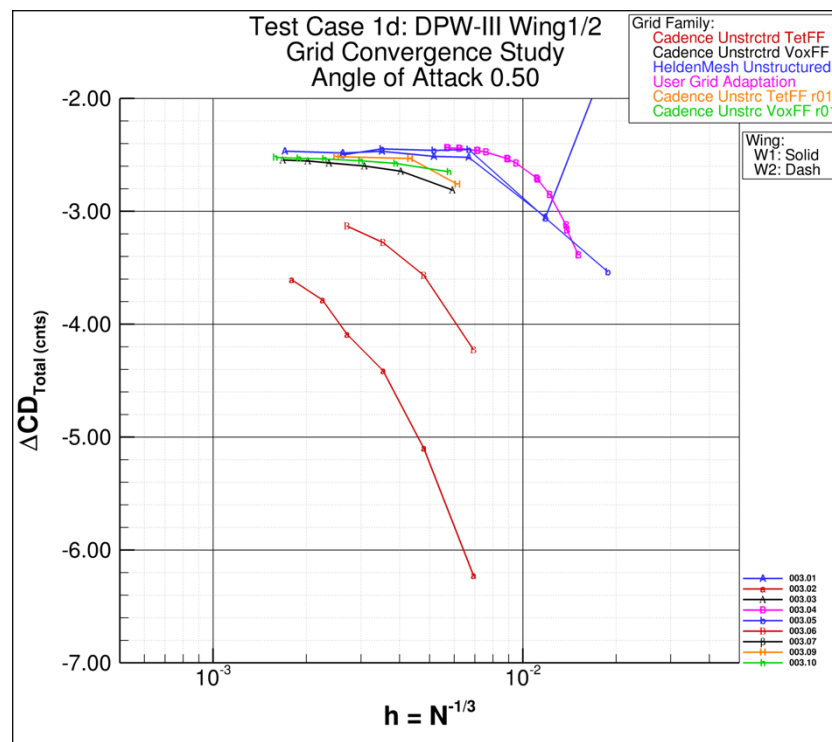


Results: Test Case 1d

• DPW-III: Wing-Only Increment Test Case

- Tet farfield REV00 (red) increments deviate!
- Can not assume increments are always good

Grids	OVERFLOW	FElight	GGNS
Cadence_HexTrex_TetFF.REV00		X	X
Cadence_HexTrex_VoxelFF.REV00		X	X
HeldenMesh_Grids.REV00		X	X
<i>Custom Grid Adaptation</i>			X
Cadence_HexTrex_TetFF.REV01		X	
Cadence_HexTrex_VoxelFF.REV01		X	

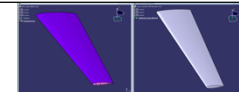
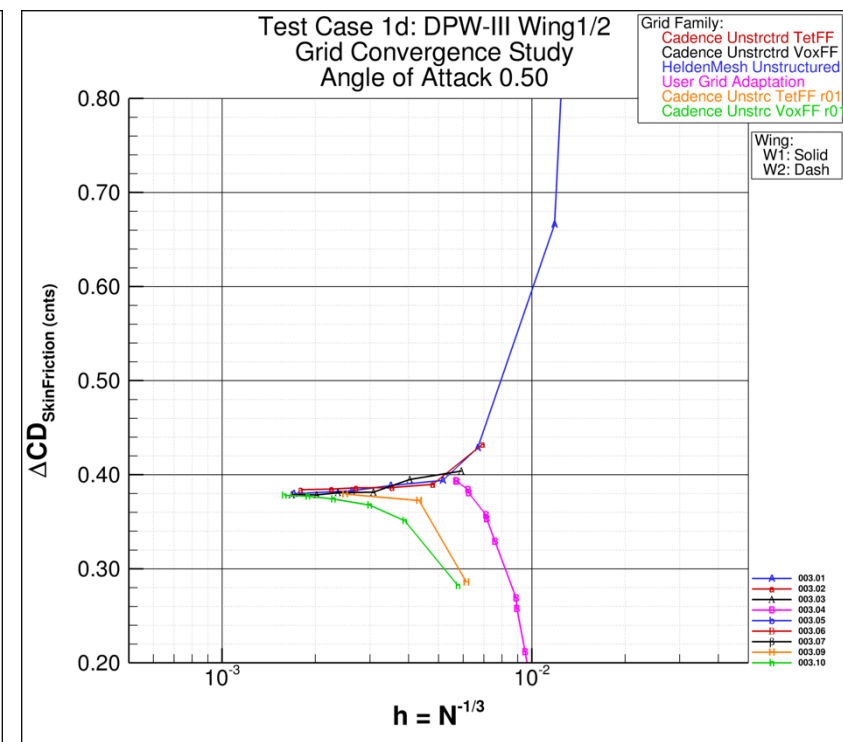
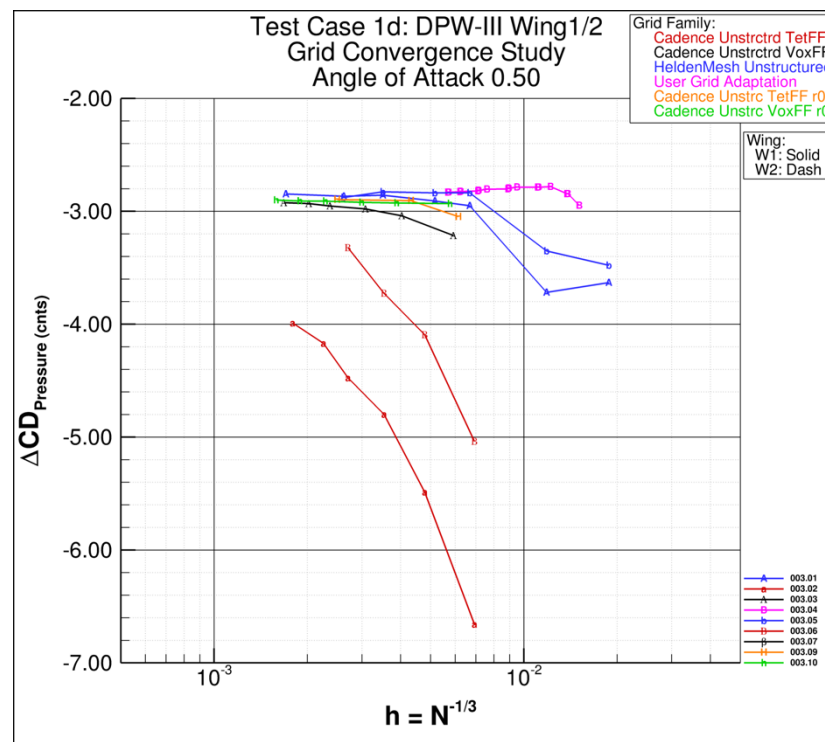
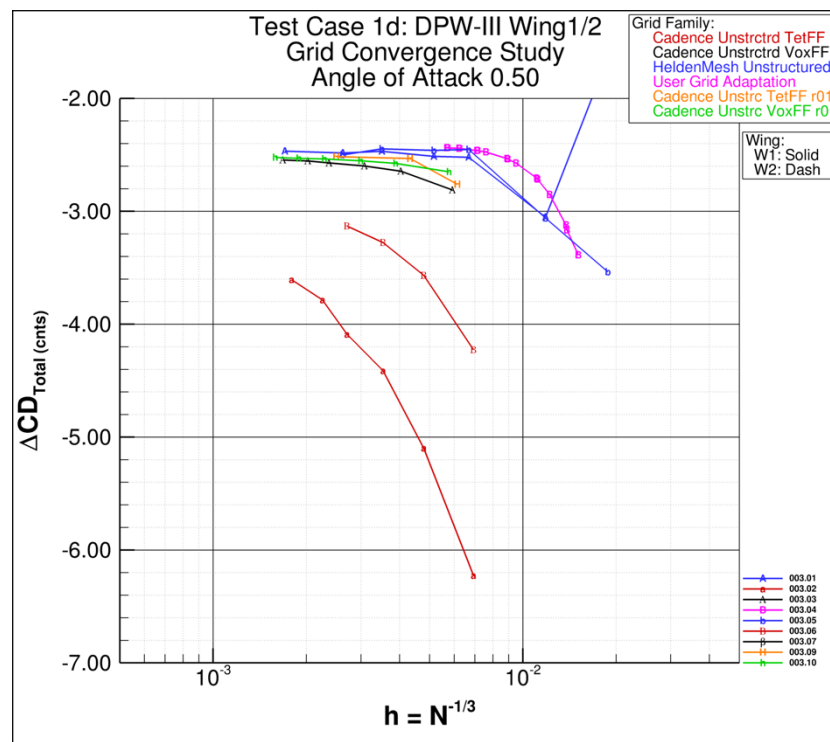




Results: Test Case 1d

- **DPW-III: Wing-Only Increment Test Case**
 - Tet farfield REV00 (red) increments deviate
 - Pressure drag is the source of the difference
 - Skin friction matches well

Grids	OVERFLOW	FELight	GGNS
Cadence_HexTrex_TetFF.REV00		X	X
Cadence_HexTrex_VoxelFF.REV00		X	X
HeldenMesh_Grids.REV00		X	X
<i>Custom Grid Adaptation</i>			X
Cadence_HexTrex_TetFF.REV01		X	
Cadence_HexTrex_VoxelFF.REV01		X	



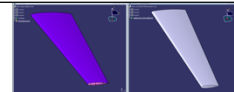
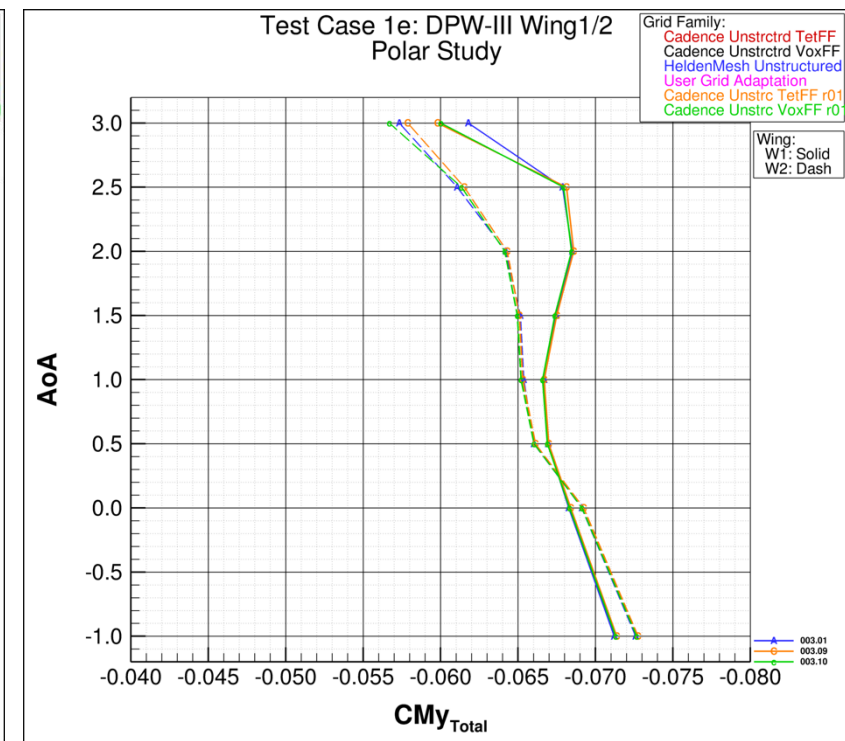
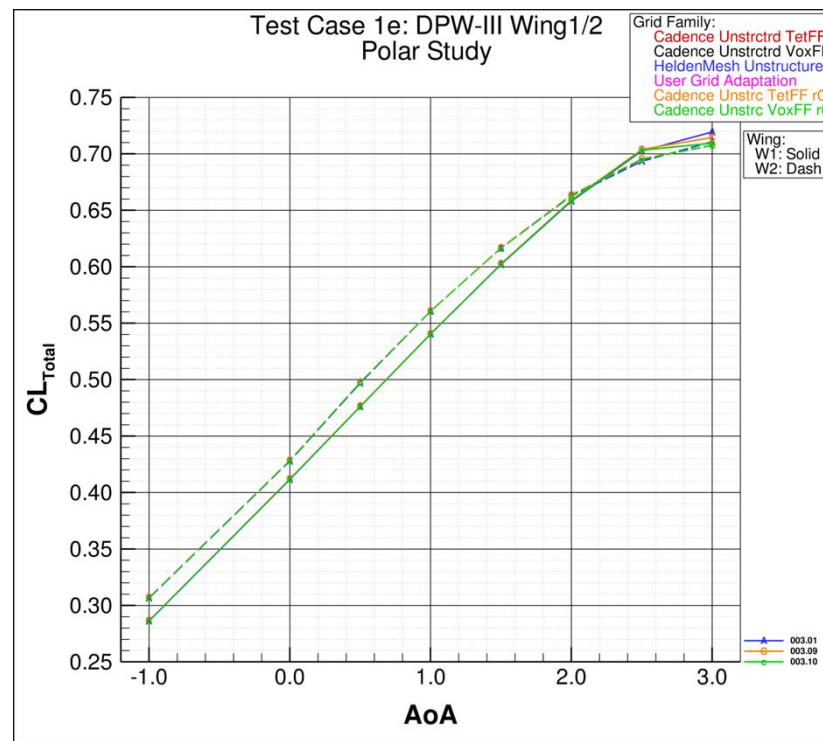
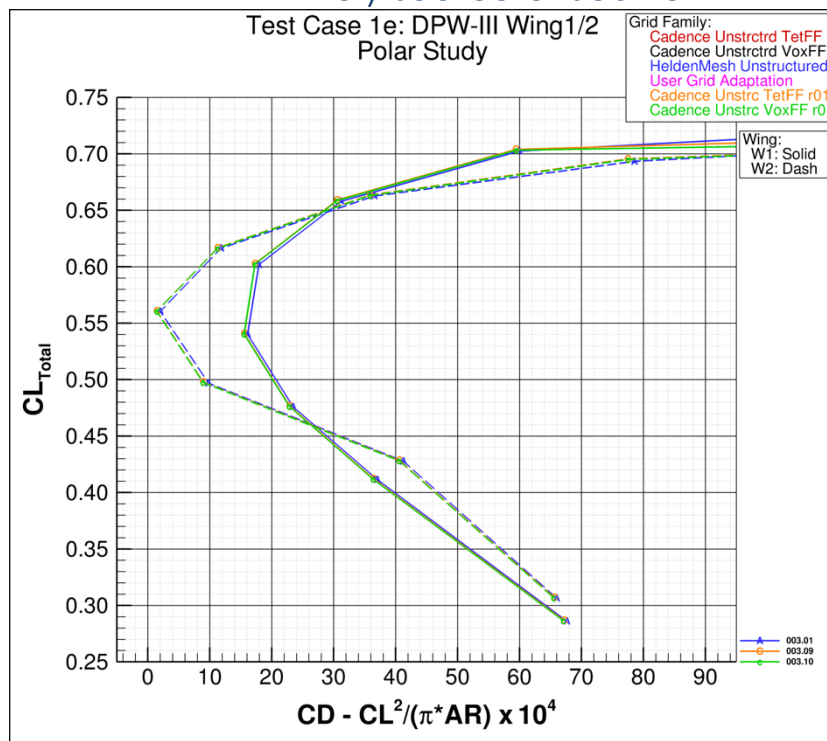


Results: Test Case 1e

• DPW-III: Wing-Only Test Case

- Good agreement between three meshes for $\alpha = [-1.0, 2.5]$
 - Drag scatter less than +/- 0.5 counts
- Residual convergence very difficult for $\alpha = 3.0$
 - Likely source of scatter

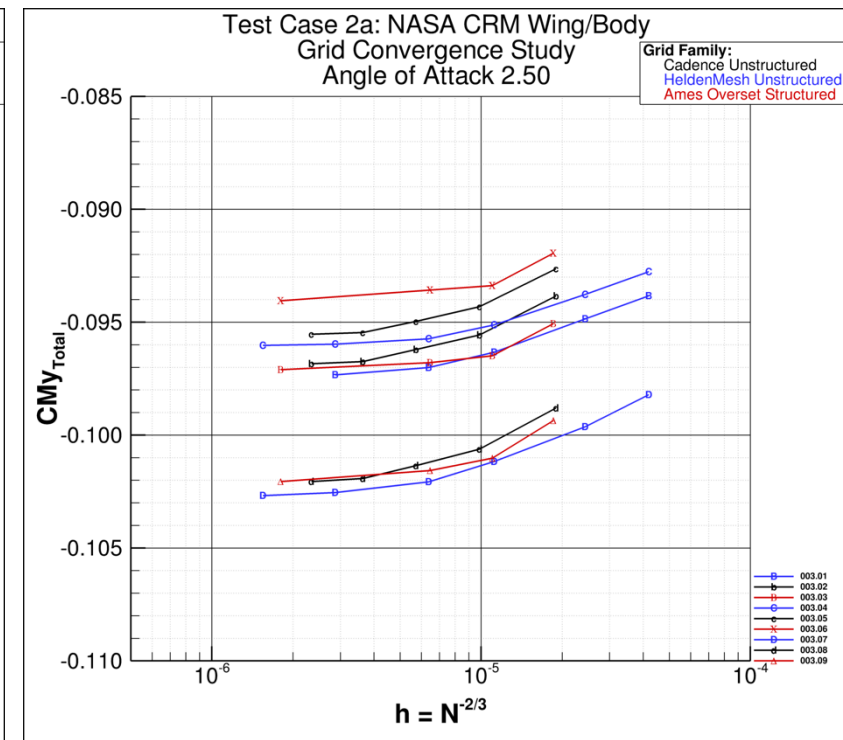
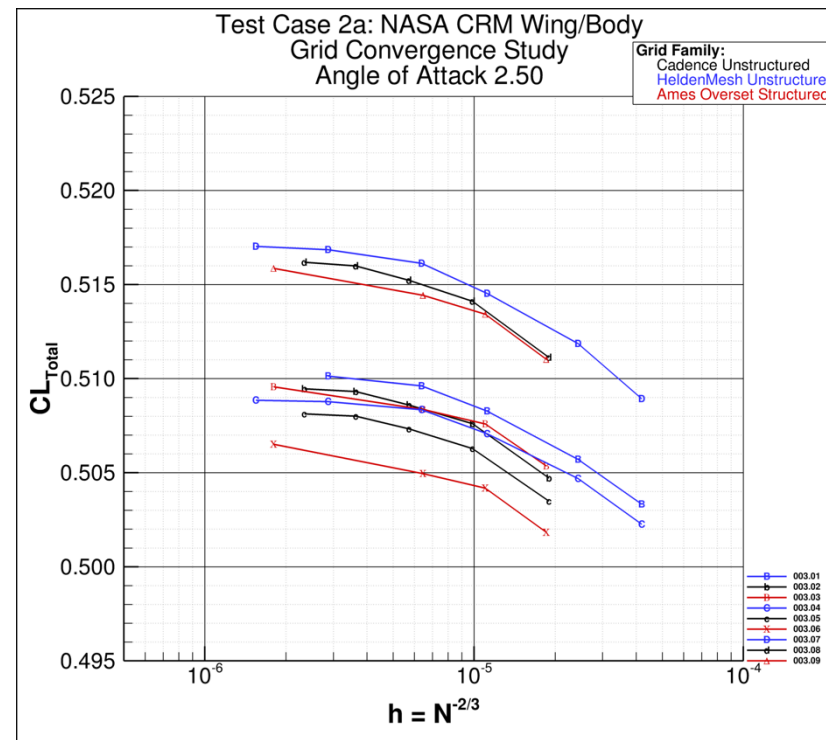
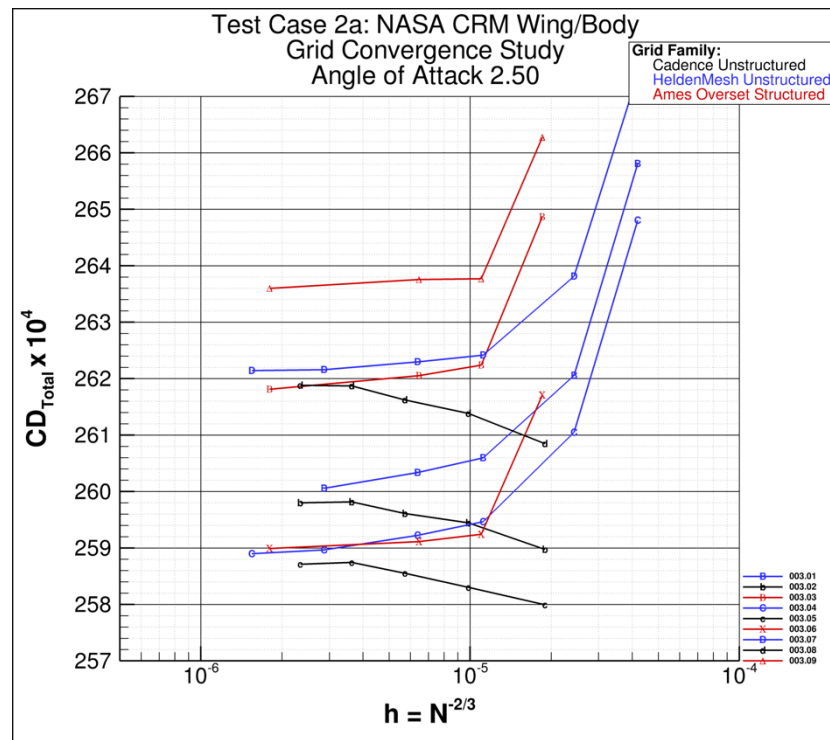
Grids	OVERFLOW	FElight	GGNS
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Cadence_HexTrex_VoxeIFF.REV00		X	X
HeldenMesh_Grids.REV00		X	X
<i>Custom Grid Adaptation</i>			X
Cadence_HexTrex_TetFF.REV01		X	
Cadence_HexTrex_VoxeIFF.REV01		X	



• NASA CRM Wing/Body

- Lines colored by grid family
- Grid convergence not yet demonstrated
- Drag scatter $\sim \pm 2.5$ counts

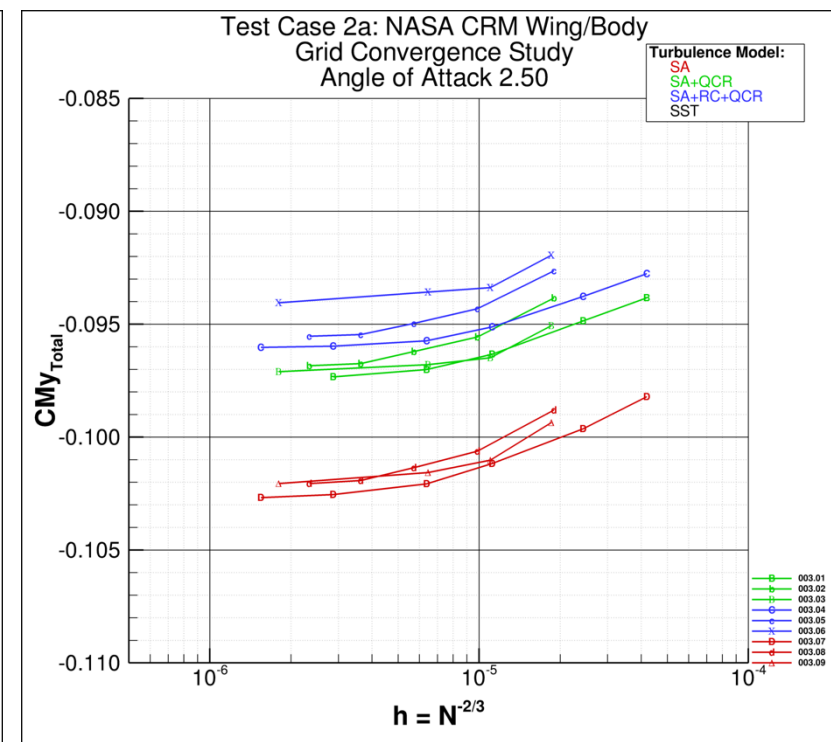
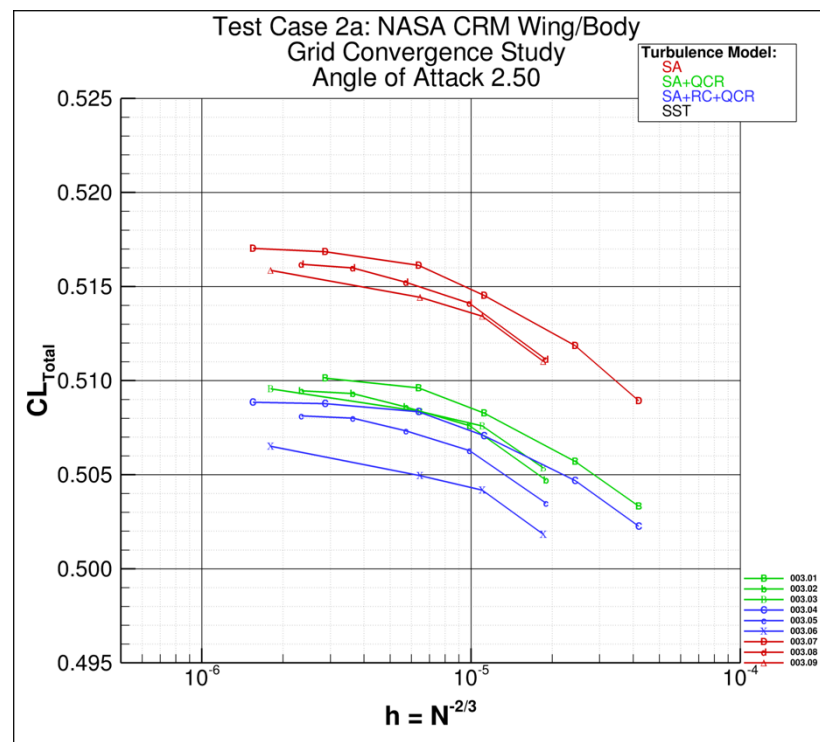
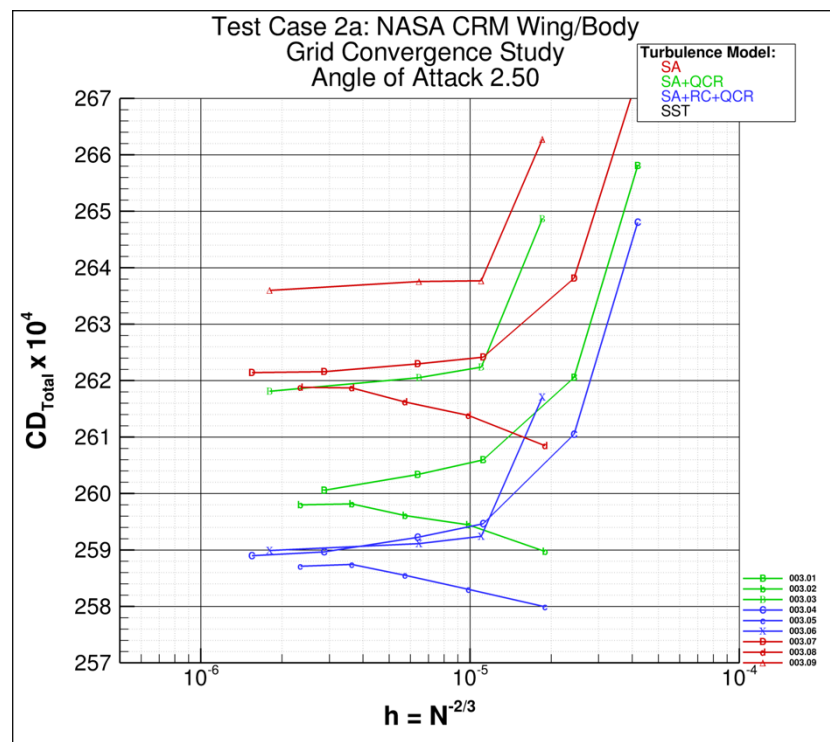
Grids	OVERFLOW	FELight	GGNS
Cadence_Grids.REV00		X	
HeldenMesh_Grids.REV00		X	
Ames_Grids.REV00	X		



• NASA CRM Wing/Body

- Lines colored by turbulence model
- RC correction to SA model shows clear impact
 - Drag scatter $\sim \pm 1$ count within the same turbulence model

Grids	OVERFLOW	FELight	GGNS
Cadence_Grids.REV00		X	
HeldenMesh_Grids.REV00		X	
Ames_Grids.REV00	X		

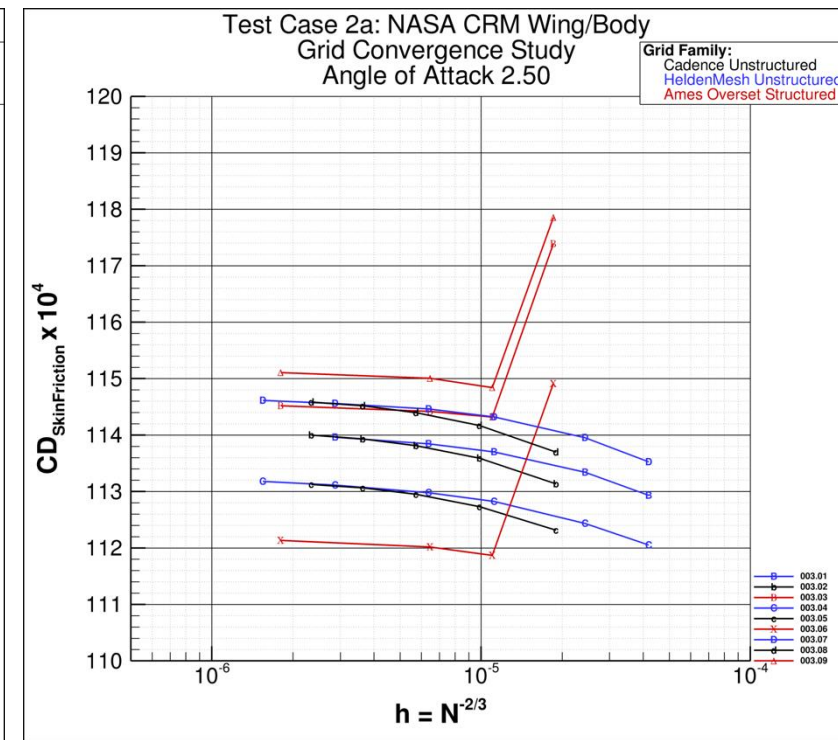
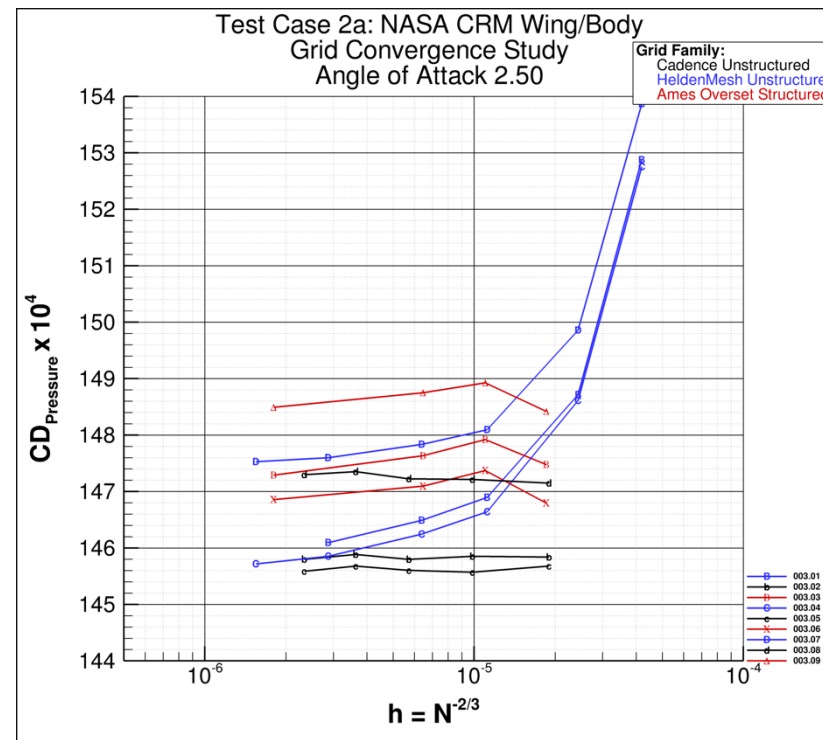
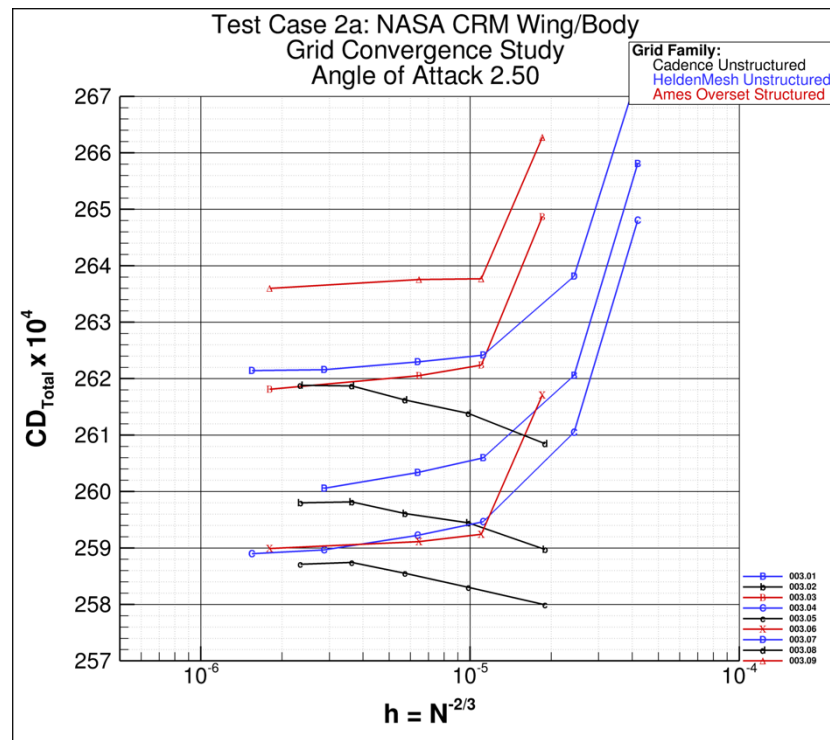


Results: Test Case 2a

• NASA CRM Wing/Body

- Lines colored by grid family
- Grid convergence not yet demonstrated
- Drag scatter $\sim \pm 2.5$ count

Grids	OVERFLOW	FELight	GGNS
Cadence_Grids.REV00		X	
HeldenMesh_Grids.REV00		X	
Ames_Grids.REV00	X		

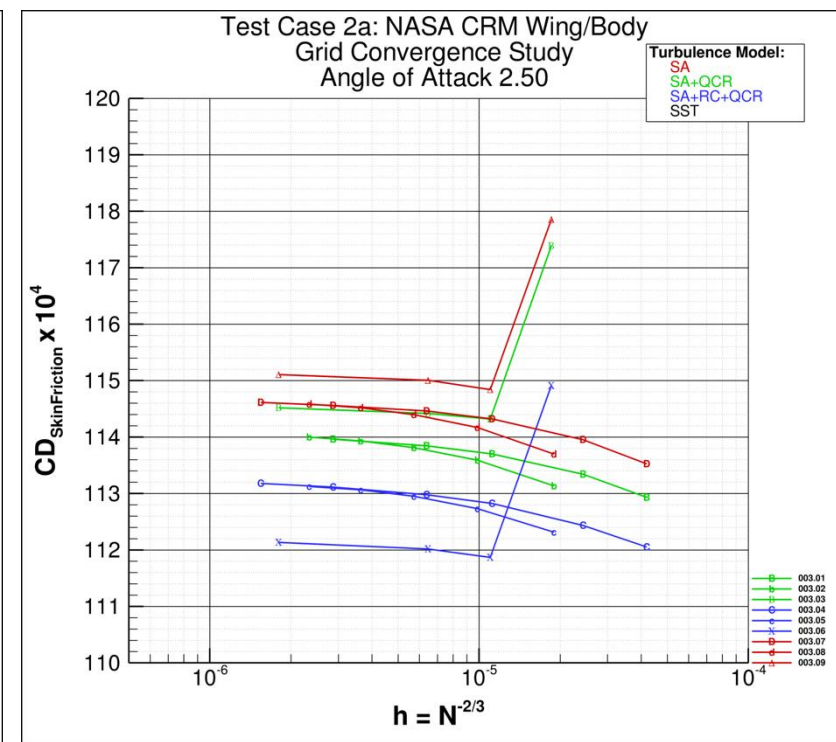
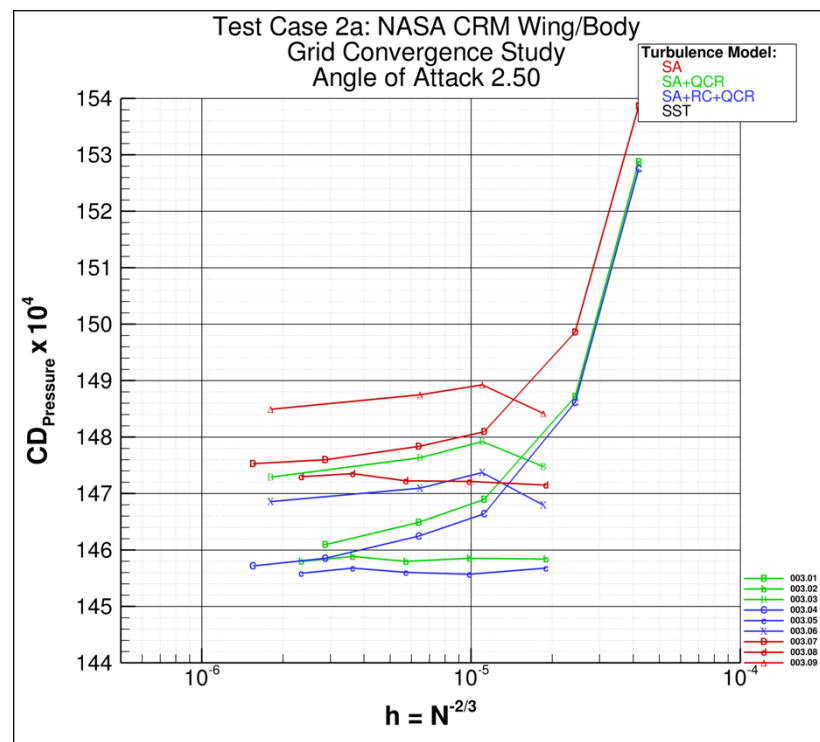
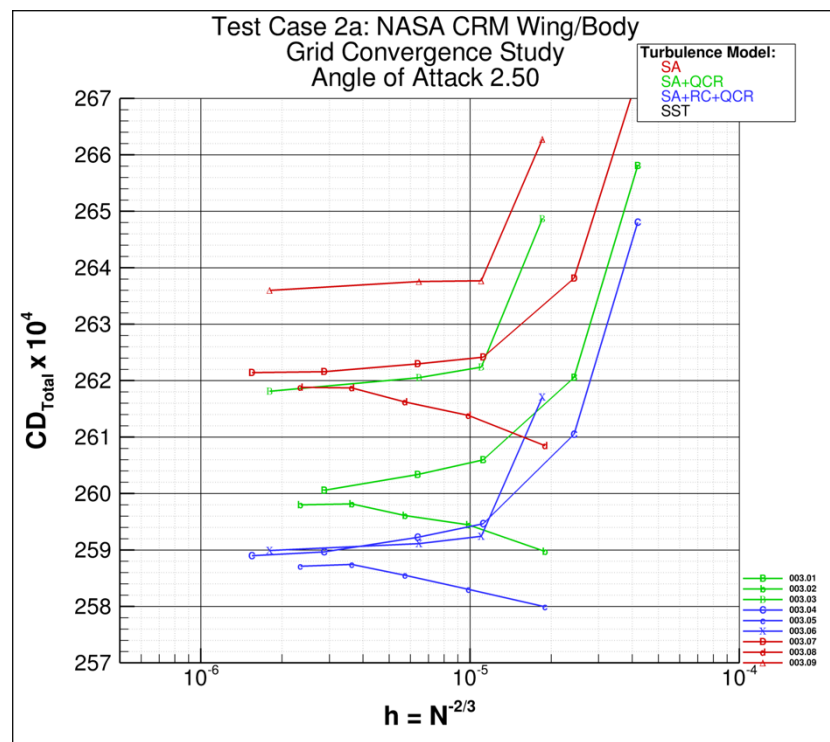


Results: Test Case 2a

• NASA CRM Wing/Body

- Lines colored by turbulence model
- RC correction to SA model shows clear impact
 - Drag scatter $\sim \pm 1$ count within the same turbulence model

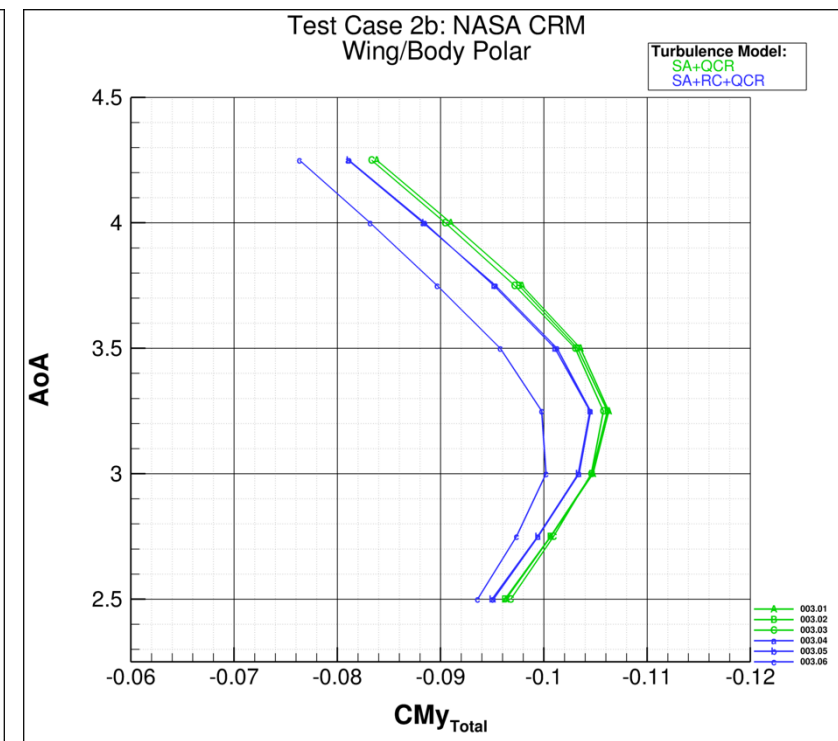
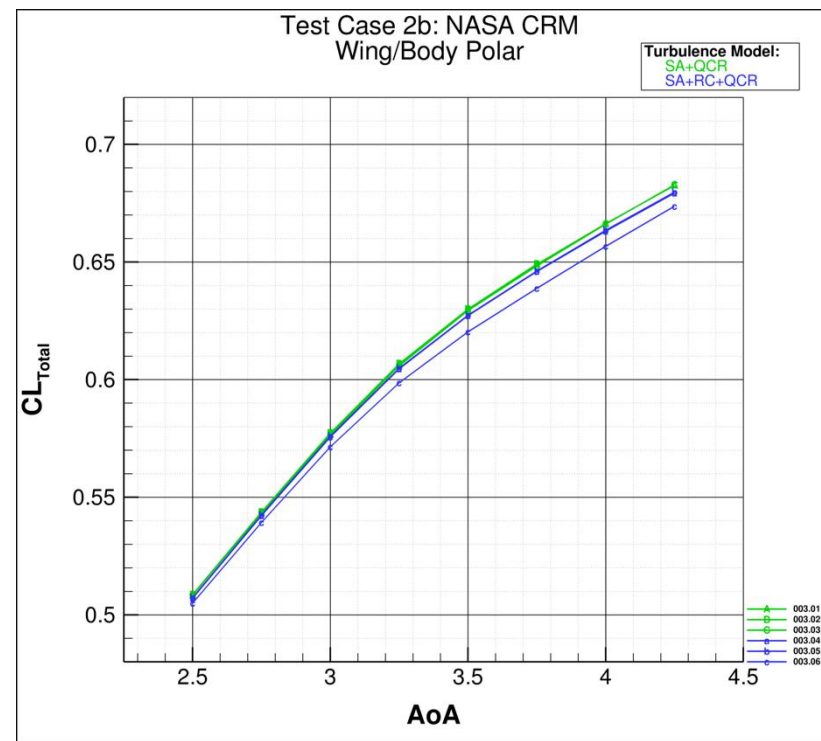
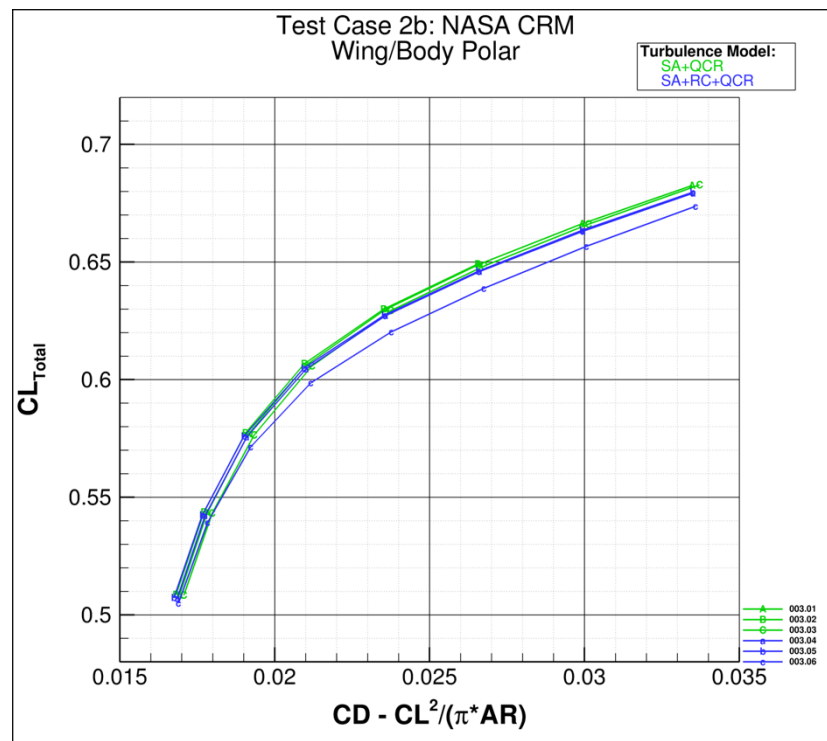
Grids	OVERFLOW	FELight	GGNS
Cadence_Grids.REV00		X	
HeldenMesh_Grids.REV00		X	
Ames_Grids.REV00	X		



• NASA CRM Wing/Body Polar

- Lines colored by turbulence model
- RC correction to SA model shows clear impact
 - Size of impact varies with solver

Grids	OVERFLOW	FELight	GGNS
Cadence_Grids.REV00		X	
HeldenMesh_Grids.REV00		X	
Ames_Grids.REV00	X		





Conclusions



- **Solver Comparison**
 - The agreement between very different grid and solver strategies is encouraging
- **Turbulence Model**
 - RC terms have an impact on forces/moments
 - Reduced lift and more positive pitching moment at constant alpha
- **Scatter Reduction**
 - Scatter reduction is possible
 - Requires diligent attention to details
 - Problem setup (all inputs, including “defaults”)
 - Grid (refinement and farfield extent)
 - Residual convergence is important



Acknowledgements



- Thank you to my colleagues who contributed data, provided support in running these cases, or provided very helpful discussion and feedback.
- Many Thanks to my organization for prioritizing our involvement in DPW efforts!

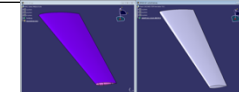
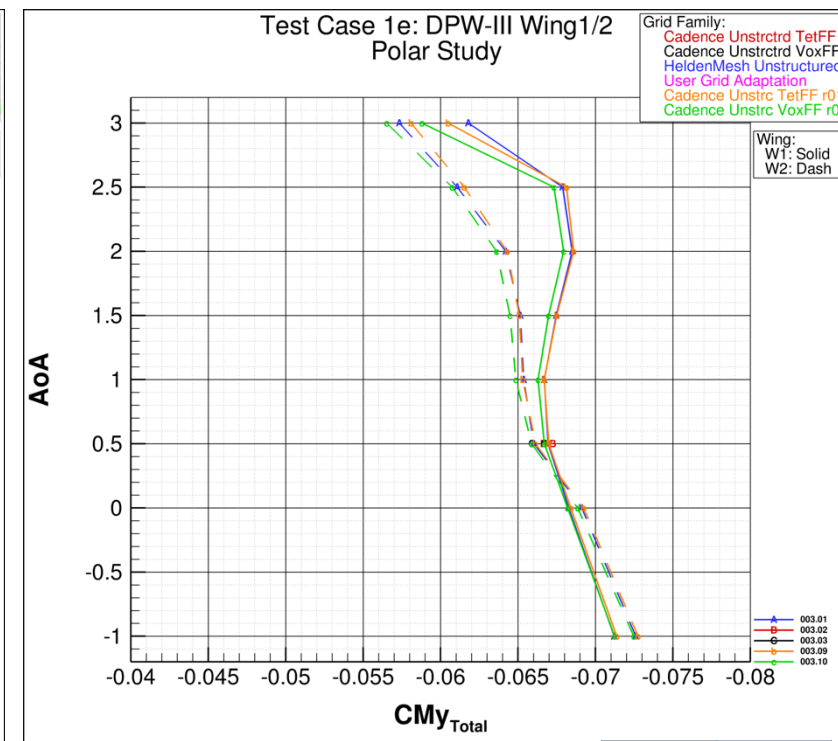
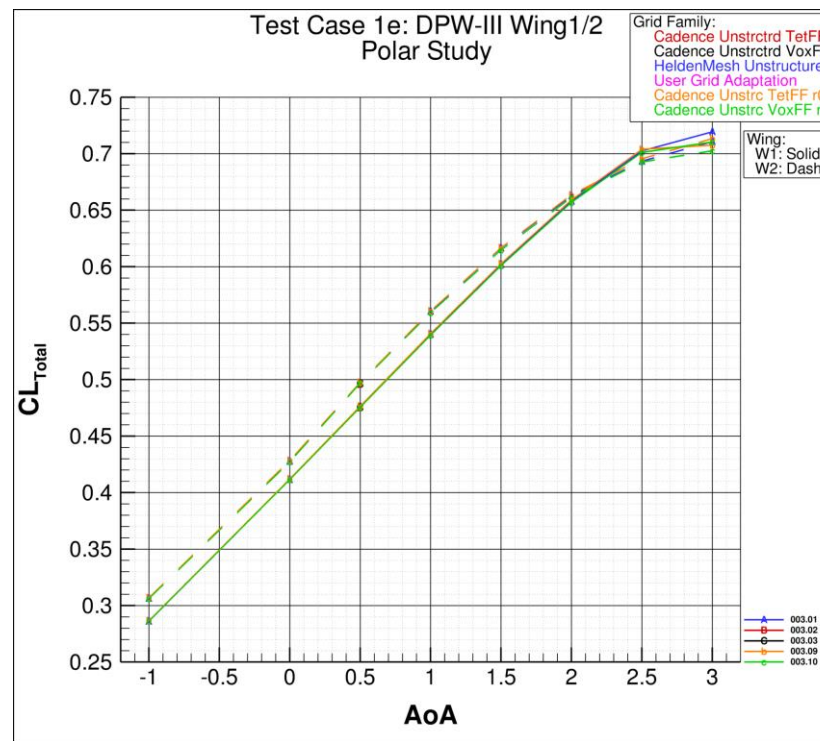
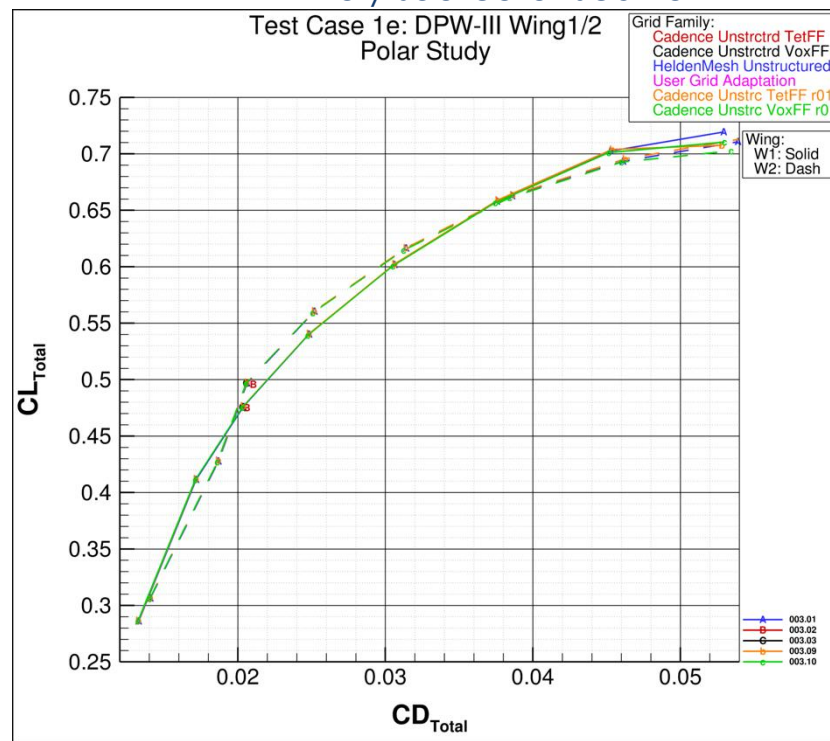


Results: Test Case 1e

• DPW-III: Wing-Only Increment Test Case

- Good agreement between three meshes for $\alpha = [-1.0, 2.5]$
 - Drag scatter less than +/- 0.5 counts
- Residual convergence very difficult for $\alpha = 3.0$
 - Likely source of scatter

Grids	FELight	GGNS
Cadence_HexTrex_TetFF.REV00	X	X
Cadence_HexTrex_VoxelFF.REV00	X	X
HeldenMesh_Grids.REV00	X	X
Custom Grid Adaptation		X
Cadence_HexTrex_TetFF.REV01	X	
Cadence_HexTrex_VoxelFF.REV01	X	





GGNS-T1

- **Discretization**

- SU/PG(1)
 - Tetrahedral cells
 - piecewise-linear, globally continuous finite elements
 - Residual-based stabilization
- Shock capturing/artificial dissipation by (Glasby et.al.[1])

- **Solver**

- RANS, steady-state (SA, SA-QCR2000, SARC-QCR2000 fully turbulent)
- Fully coupled turbulence equations
- Exact Jacobians (except SARC) with operator overloading
- Pseudo-transient continuation to steady-state
- Newton's method
- Parallel MPI implementation based on PETSc [2]
- 12 orders for the dual-volume weighted discrete residuals
- 10 orders for the adjoint linear system residual

- **Adaptation : EPIC [3]**

- Goal oriented (GO) " $\epsilon_1 + \epsilon_2$, multi-component" from [4]